Graduating into Disruption: Labor Market Outcomes for AI-Exposed College Majors*

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Abstract

How are new college graduates affected by the rise of artificial intelligence, and what can this tell us about the mechanisms behind AI's overall labor market effects? We use administrative records on college graduates to observe how economic outcomes among college majors with differing levels of labor market AI exposure have evolved since large language models became available. We find that post-graduation employment, earnings, and job switching patterns among the most AI-exposed college majors began to diverge immediately following the introduction of ChatGPT in late 2022. In regression-adjusted estimates, the most AI-exposed decile of college majors saw their likelihood of initial employment decline by five percentage points, while full-quarter initial earnings declined by thirteen percent. This earnings decline is comparable in magnitude to the earnings losses associated with graduating into a large recession. Roughly half of the decline in earnings is attributable to lower earnings within the industry sectors that employ these graduates, with the remainder resulting from a shift in the industry mix into lower-wage sectors such as restaurants and retail. The effects attenuate as graduates move further from labor market entry but remain substantial for the most exposed majors.

Keywords: Artificial Intelligence, College major, Postsecondary employment outcomes

JEL classification: J23, J24, I26, O33

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1 Introduction

Over the past several years, rapid advances in large language models (LLMs) and generative artificial intelligence have substantially expanded the set of tasks that software can perform autonomously. Unlike earlier waves of automation, such as industrial robotics or routine-task computerization, which primarily substituted for predictable manual or clerical activities (e.g., Autor and Dorn, 2013), generative AI extends automation into information processing, communication, and other non-routine cognitive activities that are central to many high-skill and college-intensive occupations. This development has triggered widespread and increasing adoption across firms (Bonney et al., 2024; Bick et al., 2026), raising fundamental questions about how labor demand, wages, and occupational structure respond when a general-purpose technology augments or substitutes for cognitive labor. Understanding these effects is therefore central to evaluating the distributional and aggregate consequences of AI-driven technological change in the labor market.

These questions are particularly salient for new college graduates. Recent analyses have found evidence of sizable effects of AI among young workers at the occupation and industry levels (Brynjolfsson et al., 2025; Hosseini Maasoum and Lichtinger, 2025; Tucker, 2026). Focusing on graduates allows us to study these effects at the clearly-defined point of labor market entry. This is an important margin because labor market conditions at graduation have also been shown to affect initial employment and earnings and to influence longer-run career trajectories (Oreopoulos et al., 2012; Altonji et al., 2016; Haltiwanger et al., 2018; Scott-Clayton et al., 2025). These findings suggest that the rise of AI could have long-term consequences for early career workers, especially recent college graduates who have made costly human capital investments in AI-exposed fields.

Recent college graduates also offer several identification advantages for studying these effects. With limited prior work experience, much of recent graduates' accumulated human capital is observable through their college education, particularly their field of study. AI exposure is typically defined at the task or occupation level. College major provides a natural way to characterize the tasks and skills that graduates are likely to supply to the labor market and, through the occupations that use those skills, their exposure to changes in labor demand. Most importantly, major is determined before post-college employment outcomes are observed. Defining AI exposure by major therefore allows us to follow graduates through employment, nonemployment, and changes in occupation and industry without defining their exposure using the job they ultimately obtain. This is particularly useful in a setting where changes in demand for AI-exposed skills may themselves affect whether and where graduates become employed.

In this paper, we provide novel evidence on the effects of AI on new college graduates by analyzing the post-graduation labor market outcomes of bachelor's degree graduates from a subset of institutions in the Census Bureau's Post-Secondary Employment Outcomes (PSEO) data, covering roughly 29% of U.S. bachelor's degrees conferred from 2016 to 2024. We first construct a measure of AI exposure at the level of college major by combining task-level exposure scores with the occupational composition of recent graduates observed in the American Community Survey (ACS). For each major, we recover the distribution of occupations for young college graduates and map these occupations to task-based AI exposure measures from Eloundou et al. (2024). This produces a pre-ChatGPT measure of the AI exposure associated with the labor market opportunities of graduates from each major. We then link these graduates to their labor market histories in the Longitudinal Employer-Household Dynamics (LEHD) data and compare employment and earnings outcomes across majors with different levels of AI exposure around the widespread introduction of LLMs in late 2022. Our event-time design estimates whether graduates from more exposed fields experienced differential labor market outcomes following the introduction of these technologies relative to their fellow graduates in less-exposed majors.

We find that graduates in the top decile of AI-exposed majors experienced a five-percentage point decline in their likelihood of employment in the quarter after graduation in comparison to graduates in the bottom six deciles of AI exposure. They also saw declines in associated quarterly earnings of about thirteen percent relative to the least exposed fields, attenuating to about five percent after two years. Taking into account the role of partial employment during a quarter, our estimates for the most exposed graduates are comparable in magnitude to the effects of entering the labor market during a large recession (Altonji et al., 2016; Oreopoulos et al., 2012). The employment rate of new college graduates remains high even at short time horizons, reflecting the typical pattern of quick transition into the labor market. However, we find that negative changes to these patterns are concentrated in the top decile of majors, which are composed largely of computer science, information systems, and other degree fields closely related to software development.

Employment and earnings are two important margins along which we can measure the impact of AI on new graduates, but neither outcome alone reveals how graduates adjust. Our matched employer-employee framework and longitudinal design allow us to observe where graduates become employed, and identify a quantitatively important margin of worker adjustment. We show that about half of the earnings decline among graduates in the most AI-exposed decile comes from changes in the sectoral composition of employment. Following the introduction of ChatGPT, the most AI-exposed college graduates saw their employment shift substantially away from high-paying industry sectors such as Information and Professional, Scientific and Technical Services, and toward lower-paying industries such as Retail Trade and Accommodation and Food Services. In comparison, the industry composition of less-exposed graduates changed little over the same period. This sectoral reallocation is consistent with greater underemployment among exposed graduates early in their careers.[1] Increased job switching rates and attenuated earnings effects at longer horizons are consistent with subsequent adjustment as these workers move from their initial jobs toward higher-paying employment. However, given the recency of ChatGPT's release and the limited time series, we cannot yet assess when or whether exposed graduates will catch up to their peers who graduated before LLMs became widely available.

Our results also help to reconcile two seemingly incongruous findings in the emerging literature. Studies of hiring and employment in AI-exposed occupations and industries find substantial declines among young workers (Brynjolfsson et al., 2025; Hosseini Maasoum and Lichtinger, 2025; Tucker, 2026). Other studies find little evidence that workers previously employed in highly exposed occupations have experienced greater increases in unemployment (Eckhardt and Gold-

[1] Abel and Deitz (2016) develop a measure of underemployment based on jobs that typically do not require a college degree. We do not develop such a measure, but the shifts we observe are toward sectors with lower college-educated employment (U.S. Census Bureau, 2025).

schlag, 2025; Massenkoff and McCrory, 2026). Research designs like these necessarily condition on employment and assign exposure using an occupation or industry that is itself an *ex post* labor market outcome. This makes it difficult to capture adjustment that occurs through delayed employment or sorting into different occupations and sectors. Our results show that these margins are important. Graduates from the most exposed majors remain strongly attached to the labor market, but they adjust to weaker demand at entry. They are less likely to find employment immediately after graduation and, when employed, are more likely to begin their careers in lower-paying sectors.

Reduced employment among AI-exposed workers could arise through several mechanisms, of which direct substitution of skilled labor is only one. We therefore examine alternative explanations for the differential outcomes we observe. Our results are not readily explained by a pure increase in graduate supply. Although the supply of graduates in the top exposure decile rose sharply before 2022, employment kept pace until the post-ChatGPT period, when it fell below trend even as supply growth slowed. Our estimates are robust to controlling for differences across majors in their work-from-home intensity, a prominent alternative explanation for the employment declines associated with AI exposure (Lambert and Schindler, 2026). Our estimates are also robust to alternative measures of AI exposure (e.g., the usage-based measure of Handa et al., 2025). Additionally, although increases in self-employment or graduate school attainment might explain a substantial portion of declines in initial employment, we find evidence that these patterns may be more consistent with weak labor demand than with increased opportunity.

The remainder of the paper proceeds as follows. Section 2 describes the PSEO, ACS, and LEHD data and the construction of our major-level AI exposure measure. Section 3 presents the empirical strategy. Section 4 reports the main employment, earnings, industry, and job-switching results and examines how they evolve with time since graduation. Sections 5 and 6 examine heterogeneity across majors and variation in the distribution of occupation-level exposure. Section 7 discusses the mechanisms and broader implications of the findings, and Section 8 concludes.

2 Data sources

2.1 Graduate data

Our sample of graduates comes from the U.S. Census Bureau's Post-Secondary Employment Outcomes (PSEO) data product. PSEO is constructed from a collection of graduation and transcript records voluntarily provided by postsecondary institutions and state higher education systems across the United States. These records contain the date of graduation, degree-conferring institution identified by Office of Postsecondary Education ID, and declared majors for every graduate.[2] For this research, we utilize the roster of graduates from twenty-seven data partners across twenty-four states representing over 350 institutions. While additional partners exist in the PSEO partnership and consented to have their data included, we restricted our sample to institutions with at least one bachelor's degree graduate in every year between 2012 and 2024.[3] Appendix Table A1 contains the list of participating systems.

We make three sample restrictions from the graduate sample. First, we exclude students with three or more majors. Double majors have the combination of their two majors treated as a single unique major. For instance, someone who double majors in "Accounting" and "Finance" is treated as majoring in "Accounting-Finance". Second, we exclude students who graduated from a college that does not appear in our data in every year from 2012 to 2024. This removes colleges that closed or opened during our sample period and colleges that have not provided up-to-date data to PSEO. Finally, we exclude students who majored in a field that only one college produced graduates in. These observations would be automatically dropped in our fixed effects models involving major-college dummy variables, but we drop them at the onset to exclude them from all analyses. Our filtered sample consists of 6,665,500 graduate records, capturing approximately 29% of all bachelor's degrees conferred from 2016 to 2024.[4]

[2] The PSEO data categorize majors by their Classification of Instructional Program code as defined by the U.S. Department of Education. We crosswalk these codes to the fields of degree utilized by the Census Bureau for the American Community Survey.

[3] In future work, we hope to include degree levels beyond the bachelor's level; however, data limitations force us to restrict our attention to bachelor's graduates at this time.

[4] All statistics in this paper other than those derived solely from publicly available data are rounded according to

Appendix Table A2 presents summary statistics for institutions in our sample based on publicly available data in the Integrated Postsecondary Education Data System (IPEDS) for academic year 2022. Relative to the broader postsecondary landscape in the United States, our set of institutions are significantly more likely to be public as opposed to private or for-profit. They are also more likely to be located in cities as opposed to suburbs, towns, or rural areas, have enrollments above 20,000 students, and be classified as doctoral-degree granting with either "very high" or "high" research output. These institutions also graduate students that are slightly more likely to be male and white (non-Hispanic). Overall, these differences are similar to those documented by Orr (2025) for the entire PSEO sample, not just the set of institutions participating in this particular research. Nonetheless, if the effects of AI are heterogeneous by institution type, then our estimates of AI's impacts may only be representative of this subset of institutions, and not the full postsecondary landscape.[5]

2.2 GenAI Exposure

There are multiple measures of AI exposure in the nascent literature, but the measure developed in Eloundou et al. (2024) has become a widely used benchmark. Eloundou et al. use the GPT-4 large language model to classify detailed work activities and tasks in the Department of Labor's occupation database (O*NET) as either AI-exposed or not. They determine exposure by asking whether GPT-4 can reduce the time it takes to complete the task by at least half, giving a full point if yes and half a point if GPT-4 could reduce the time by half if it had access to additional software. Occupational exposure measures are the sum of task exposures. Their preferred weighting scheme, known as GPT-4 Beta, gives core tasks (as indicated in O*NET) twice the weight as supplemental tasks. The GPT-4 Beta measure is used as a primary measure of occupational exposure in several recent analyses of employment and hiring (Brynjolfsson et al., 2025; Chandar, 2025; Hyman et al.,

the U.S. Census Bureau's disclosure avoidance rules.

[5]Note that our baseline regression specifications all include institution-by-major fixed effects, so we identify outcomes from variation over time within each institution and major rather than from variation across the composition of institutions. Future work will examine heterogeneity in outcomes across the different types of institutions we can observe.

2025; Johnston and Makridis, 2026; Tucker, 2026) and has been used to discipline quantitative analyses of AI's labor market and aggregate effects (Acemoglu, 2025; Freund and Mann, 2026). The measure is also informative about realized adoption. Bick et al. (2026) find that predicted occupational exposure is positively associated with individuals' reported generative AI use at work, while Tucker (2026) finds that the GPT-4 Beta measure is highly predictive of subsector-level AI utilization reported by firms in the Census Bureau's Business Trends and Outlook Survey (BTOS).[6]

The Eloundou et al. GPT-4 Beta measure is part of a class of exposure measures based on AI's capacity to perform certain tasks, but not necessarily reflective of its actual usage. In contrast, Handa et al. (2025) use an automatic classifier to analyze millions of actual prompts to Claude.ai and map them to O*NET tasks and objectives. In Appendix B, we explore whether our conclusions differ when calculating major-level exposure using several alternative measures of occupation-level exposure, including the measure of Handa et al. There is general agreement in which majors are most exposed regardless, and our results are robust to adopting an alternative measure.

AI adoption patterns continue to evolve, and the measures used in this paper predate many model innovations as well as the widespread diffusion of agentic platforms. Accordingly, we view AI exposure measures as snapshots of a rapidly changing technology.[7] The underlying extent of college majors' AI exposure may evolve as new model and platform capabilities arise. More generally, technical exposure does not translate mechanically into adoption or labor substitution. Lindenlaub et al. (2026) show that adoption depends on AI's comparative advantage relative to a particular worker, accounting for both AI user costs and worker productivity, while Lashkari et al. (2026) emphasize fixed implementation costs and the scale at which a task is performed. For now, consistent with much of the empirical literature, we treat exposure as a fixed pre-period assignment. The effects we estimate are in relation to this fixed assignment, based on deciles of exposure. The variation in our coefficients over time may relate to the changing intensity of AI capability or

[6]The GPT-4 Beta measure has the third highest predictive power for BTOS utilization in the measures examined in Tucker (2026) with the core and total measures in Eisfeldt et al. (2023) having the highest predictive power. We choose to use the GPT-4 Beta measure because it is highly correlated with the Eisfeldt et al. measure in our setting (Spearman rank correlation of 0.93 at the college major level) but enables direct comparison with more research quantifying the effects of AI. Select results using the Eisfeldt et al. exposure measure are contained in Appendix B.

[7]In general, if major-level exposure changes over time, our specifications should capture the relevant effects of AI exposure as long as these changes are rank-preserving.

changes in the labor market response to AI. We cannot necessarily distinguish between these two margins.

2.3 Occupation and major mapping

The natural unit of work classification to define AI exposure is the task, but rarely do worker-level data contain task-level records. Instead, the collection of tasks generally performed by occupations are provided by O*NET, and AI exposure is aggregated to the occupation level based on the listed tasks. One way to map the occupation-level exposure measures to college majors, as we must do in this work, is to use a distribution of realized occupations for graduates of each major to construct an occupation-weighted-average (Timmerman, 2025). Since the vast majority of state Unemployment Insurance wage records do not provide occupation, we do this with survey data. Beginning in 2009, the American Community Survey (ACS) started asking respondents with a bachelor's degree for the majors associated with their degree. In addition, the ACS asks respondents to describe the occupation of their primary job worked in the last week. To make the major-occupation distribution more relevant to our population of interest, we use ACS survey years 2018 to 2022 and restrict the sample to workers aged 21 to 29 with only a bachelor's degree (Ruggles et al., 2025).[8] With major-occupation distributions estimated, we develop our baseline AI exposure measure as the graduate-weighted mean of occupational exposure scores. To examine one possible dimension of heterogeneity in exposure, we also define a second measure based on the share of occupations worked by a major that is in the top quintile (at the occupation level) of AI exposure. This alternative measure allows us to differentiate between majors that work in occupations consistently exposed to a moderate level of AI versus majors that work in a mix of very low and highly exposed occupations. The results are presented in Section B.

In Appendix Table A3, we provide a detailed list of the AI exposure scores that we construct for each major, including our baseline GPT-4 Beta measure, as well as measures such as Handa et

[8]We choose the 21-29 age range to ensure sufficient sample size for each college major while also capturing the set of possible early career occupations that are relevant for a recent graduate. Additionally, if the career ladder faced by recent graduates involves later occupational switches to more exposed occupations, we want to capture that potential exposure.

al.'s observed AI usage and the core task exposure measure of Eisfeldt et al. (2023). Notably, the most exposed decile of college majors in our baseline measure is heavily dominated by computer science, information systems, and other majors linked to software development. However, the overall mix of majors with above-average AI exposure is much broader.

2.4 Employment data

Our labor market outcomes come from the Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) program. The LEHD data are built from state Unemployment Insurance wage records and the Quarterly Census of Employment and Wages and contain the employment status, earnings, and employer characteristics for over 95% of jobs in participating states (U.S. Bureau of Labor Statistics, 2026b). In addition, the LEHD data contain worker-level demographic and residence information obtained from a variety of administrative and survey sources.[9]

All but two states – Alaska and Michigan – have data available for every quarter between Q1 2016 and Q3 2025. To minimize classifying employed individuals from Alaska and Michigan as unemployed, and to mitigate compositional changes from Alaska and Michigan workers appearing for some years of the sample, we exclude from our sample any individuals who are *employed* in Alaska or Michigan during the analyzed time period and any individuals *residing* in Alaska or Michigan during the analyzed time period. Thus, employment rates should be interpreted precisely as the likelihood of having UI-covered employment in a state other than Alaska or Michigan among individuals currently residing in a state other than Alaska or Michigan.

The bottom-left panel of Figure 1 plots the mean log of real full-quarter earnings for the dominant job two quarters post-graduation (hereafter, initial earnings)[10]. This corresponds to graduates who were employed one-quarter post-graduation through at least three-quarters post-graduation

[9]LEHD data identify the state and firm in which each worker is employed, but typically impute an establishment assignment for each worker in the firm based on distance from place of residence and candidate establishment sizes (Tucker et al., 2024). We obtain industry of employment from the establishment-level record, which comes from QCEW. In limited cases where an establishment identifier is not imputed, we use the modal employment-weighted industry of the firm in the state.

[10]All earnings in this paper are adjusted for inflation according to the CPI-U, not seasonally adjusted series (U.S. Bureau of Labor Statistics, 2026a)

in the same job, so the effect of mid-quarter job transitions on earnings is minimized.[11] As with employment, the figure shows that initial earnings increased prior to ChatGPT's public release. We also observe a steep decline for the top quintile in the quarters following November 2022, though all graduates appear to see declining quarterly earnings in recent quarters. From 2022Q4 to 2024Q4, initial earnings fell for all groups by four to nine log points.

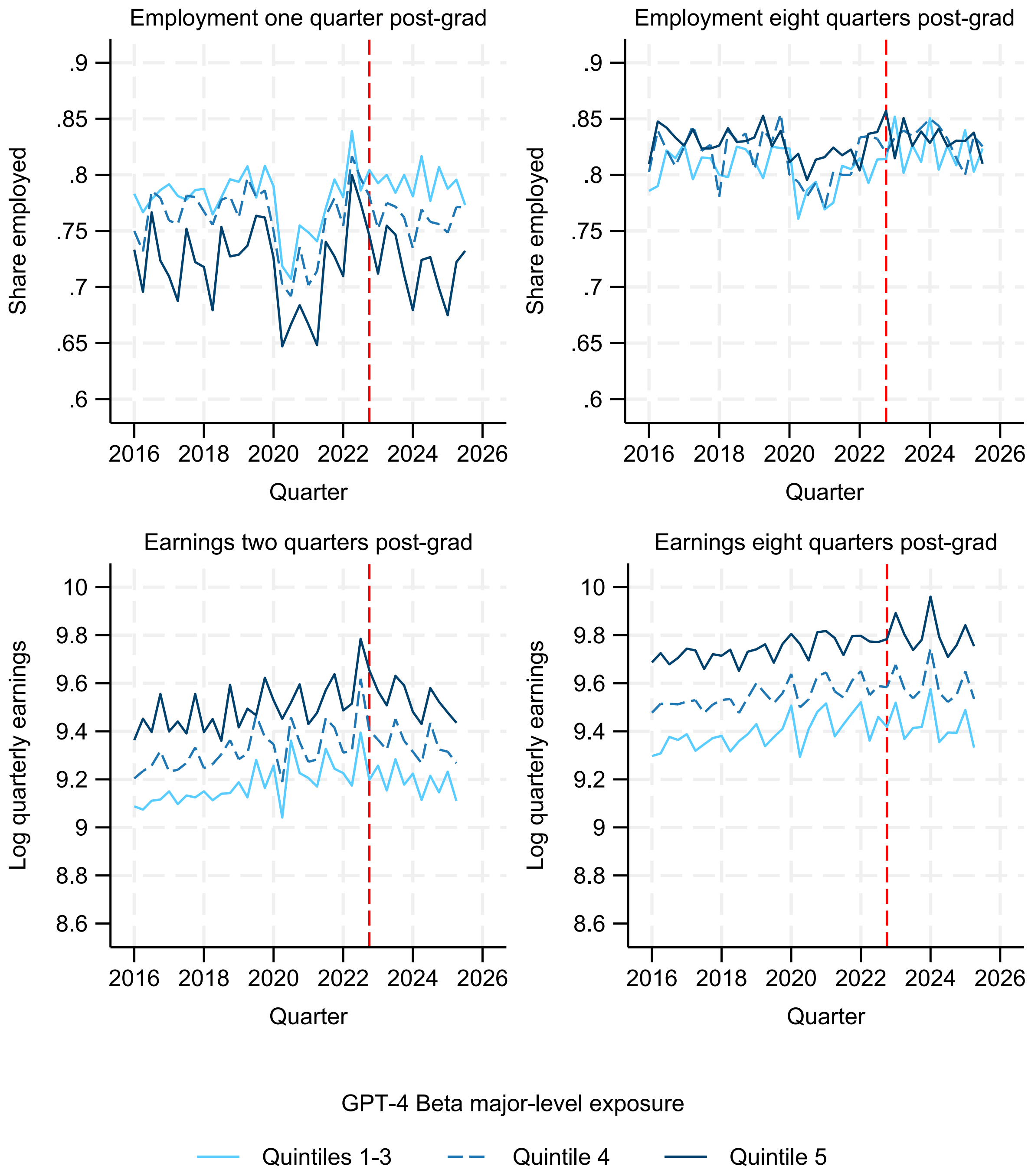
The right two panels of Figure 1 present the same outcomes for eight quarters post-graduation. Here we observe less pronounced changes over time, including pre-2022, during the COVID-19 pandemic period, and post-ChatGPT-3.5, and all quintiles of exposure appear to have fairly similar trends throughout the entire time horizon. It is important to note that the horizontal axis denotes the measurement quarter, not the graduation quarter. For instance, observations observed in 2024Q1 graduated in 2022Q1. Only the last few quarters include students who graduated post-ChatGPT, and the majority of the "post" period data points reflect students who graduated pre-ChatGPT but experienced the release of ChatGPT within two years of graduating.

We present one more piece of descriptive evidence in Figure 2 on the effects of AI before moving to our empirical strategy. Because the LEHD data are job-level files, we can track individuals over time and observe whether they changed jobs within the first two years of their post-collegiate career.[12] Job switching is an important outcome in this context because employer changes account for a substantial share of early-career earnings growth (Topel and Ward, 1992; Hahn et al., 2021), and an early switch may allow initially underemployed graduates to move toward a better job match. The trend in job switching is fairly constant prior to the COVID-19 pandemic for the two most-exposed quintiles before dropping during the pandemic period. Conversely, the least exposed quintiles have declining job switching rates in the years leading up to the COVID-19 pandemic that hasten in their decline in 2020. Following the pandemic, each exposure group saw increases in job switching that surpassed the pre-pandemic peaks, consistent with the popular con-

[11] A worker's dominant job is the job with the highest earnings during the quarter among all full-quarter jobs.

[12] We define a job switch as occurring if the graduate's first observed dominant job in the quarters strictly following graduation was with a different firm, identified by State Employer Identification Number (SEIN), than the employer of their dominant job eight quarters post-graduation. If the graduate has multiple employers in the first quarter post-graduation, we compare against the employer with the highest earnings. If a graduate has no employment eight quarters post-graduation, they are excluded from the analysis. If a graduate left their original employer but was re-hired before the two year mark, they would not be considered as having switched jobs.

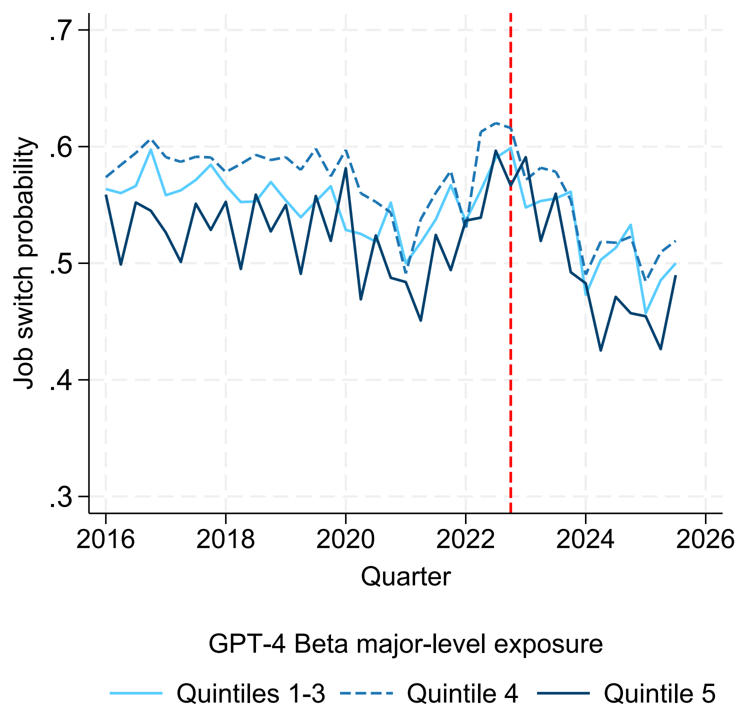
Figure 1: Employment and earnings for recent college graduates



Notes: Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. Number of observations: 6,665,500. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

ception of a "Great Resignation" over this period. Immediately after the release of ChatGPT, job switching quickly fell again. By 2025, job switching was lower for all exposure groups than their respective COVID-19 troughs, though there are signs of a reversal for those who graduated in the middle of 2023.

Figure 2: Job switching rates two years post-graduation



Notes: Job switches are identified by a change in the State Employer Identification Number (SEIN) of a graduate's dominant employer eight quarters post-graduation relative to the SEIN of the first dominant job held post-graduation. The denominator is all graduates with positive earnings in the eighth quarter post-graduation. Time labels on the x-axis are based on outcome measurement, the 8th quarter post-graduation. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. Number of observations: 4,679,050. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

The patterns shown in Figures 1 and 2 suggest that AI may be having an impact on new graduates. However, on their own, these patterns are not sufficient to conclude that exposure to AI is the cause of worsening labor market outcomes. For example, graduates from colleges that produce a disproportionately high share of exposed majors may be concentrated in geographies and industries with worsening labor prospects. Differences in institution and program quality could drive observed patterns over time if the size and composition of programs are shifting. Seasonal patterns in hiring and earnings might obscure more persistent trends. And, confounding factors like the rise of work-from-home may affect trends in ways that cannot be disentangled from the raw observational evidence alone. For these reasons, we proceed with a regression-based empirical strategy. Later, in Section 7, we extend this model to consider remote work and other alternative explanations for the patterns we observe.

3 Empirical Strategy

To address the range of concerns outlined in the previous section, we estimate the following high-dimensional fixed effects model as our baseline specification. Let y_{it} be an outcome for graduate i in time period t, $m(i)$ the major of graduate i, $E(m)$ the AI exposure for major m, d an AI exposure decile, $c(i)$ the college of graduate i, and $q(t)$ the quarter of the year for period t. Our baseline estimating equation takes the form:

$$y_{it} = \sum_{s \notin 2022} \sum_{d>6} \beta_{ds} 1[E(m(i)) \in d, t = s] + \delta_{c(i),m(i)} + \phi_{c(i),t} + \tau_{m(i),q(t)} + \varepsilon_{it}. \tag{1}$$

In this specification, $\delta_{c(i),m(i)}$ captures time-invariant factors absorbed by the interaction of college and major. For example, colleges often have specialties in certain majors that consistently perform well; the inclusion of $\delta_{c(i),m(i)}$ ensures that our results are not driven by changes in the size of these programs relative to the broader sample, only by variation within these programs. $\phi_{c(i),t}$ captures time-varying factors that affect an entire college. For example, an increase in resources for a career services center or a shock to the local labor markets into which the college's graduates typically enter would be absorbed by this fixed effect. $\tau_{m(i),q(t)}$ captures heterogeneous seasonal patterns for each major, which were shown in Figure 1 to be non-trivial. This seasonality represents both fluctuating labor market conditions and job hiring throughout the year as well as compositional changes in the types of students graduating at different times of year. Finally, each β_{ds} coefficient captures the average effect in quarter s of graduating with a major with a particular exposure to AI relative to the excluded group.

We choose to discretize AI exposure into deciles while grouping the bottom six deciles as a pooled baseline. Thus, our event study coefficients can be interpreted as the effect of AI on highly exposed majors relative to a majority of lesser exposed majors, net of the difference between these majors in the corresponding quarter in calendar year 2022.[13] This approach roughly follows the recent analyses of early-career outcomes in Brynjolfsson et al. (2025) and Tucker (2026), both of whom construct exposure quintiles and find that the effects of AI are nonlinear and concentrated in the top quintile of exposed occupations or industries. The choice to disaggregate into exposure deciles is for two reasons. First, given the size of our sample, we have sufficient power to examine nonlinearities at this level of detail. Second, because we map occupation-level AI exposure to college majors, the variation in exposure scores across majors is inherently compressed relative to that of the underlying occupation distribution. For example, the *ninetieth* percentile of exposure at the major level (0.50) is nearly the same as *sixtieth* percentile of exposure at the occupation level (0.49). In Section 5, we estimate an alternative specification for two of our outcome measures with major-specific event study coefficients net of the same set of underlying fixed effects as Equation (1). Such a specification allows us to compare several alternative measures of major-level AI exposure, including an alternative construction of exposure based on the share of occupations in the top quintile of Eloundou et al.'s GPT-4 Beta measure, shown in Section 6.

To interpret β_{ds} as a causal effect of AI exposure, the primary identifying assumption is that absent the introduction of ChatGPT, the change in outcomes from 2022 onward for AI exposed majors in deciles seven through ten would have been the same as that experienced by the bottom six deciles. Note that this may not be the full causal effect of AI exposure – we only observe outcomes for the bottom six deciles post-2022 in a world where ChatGPT is released, and it is plausible that generative AI and LLMs have an effect on students in the bottom six deciles of the exposure distribution.[14] This parallel trends assumption would fail if certain shocks occur

[13]While the underlying exposure variable is continuous, discretizing exposure and estimating a saturated model where every discrete category enters with an indicator variable preserves an average treatment on the treated interpretation (as opposed to treatment on the treated given a specific treatment dosage) without requiring an assumption of homogeneous potential outcomes (conditional on treatment dose) for all treated observations (Callaway et al., Forthcoming).

[14]As shown in A1, almost all majors have some share of their jobs in occupations that are highly exposed to AI.

that disproportionately affect majors in the top deciles of AI exposure and these shocks occur concurrently with the widespread adoption of AI tools. In Section 7, we consider and evaluate potential confounders that might threaten this assumption, most notably the rise of work-from-home, which is strongly positively correlated with AI exposure at the occupation level (Lambert and Schindler, 2026).[15]

A second assumption for a difference-in-differences estimation is no anticipation of treatment. This assumption would be violated if highly exposed graduates or employers anticipated the capabilities of ChatGPT-3.5 and subsequent releases and preemptively changed their major, job search, or hiring strategies. We believe the scope of anticipatory adjustment is small in the short window where the capabilities and public release timing of LLMs were reasonably well understood.[16] In general, violations of a no-anticipation assumption would cause us to underestimate the effects of AI on highly exposed majors as graduates try to preemptively mitigate the impacts of a weakening labor market.

Finally, difference-in-differences methods such as this one rely on a Stable Unit Treatment Variance Assumption (SUTVA). This requires independence between the potential outcome of each observation and the actual treatment status of other observations (Imbens and Rubin, 2015). This could be violated if students change their majors in response to ChatGPT's release, such as to reduce their own expected AI exposure. It may be difficult for students who are close to graduation to switch majors, but the concern becomes increasingly salient among later cohorts. Whether major switching increases or decreases measured employment and earnings outcomes likely depends on where major switchers sit in the conditional earnings distribution of both their source and destination majors.[17] More generally, we provide reduced form estimates in this paper; general equilibrium effects in the market for new college graduates could be impacting these estimates,

[15]Our analyses use 2022 as a reference period, and so we formally only require parallel trends from 2022 onward, not in the pre-pandemic or pandemic periods. However, we provide estimates from 2016 onward because these periods help to identify fixed effects and are informative of the relationship between employment and earnings of these major groups over a longer period.

[16]Notably, Eisfeldt et al. (2023) find little evidence of anticipation in stock market valuations or other measures of behavior among AI-exposed firms. Furthermore, college graduates have limited ability to change their major or graduation timing when they are close to completion.

[17]The graduate data on which we rely contains some information about major switching, and we intend to evaluate the magnitude and direction of major switching in future work.

such as by affecting market wages in the occupations that employ both more- and less-exposed college graduates. These general equilibrium effects would also be a violation of SUTVA, but it is not clear *ex ante* what direction such effects would bias our results.

4 Results

4.1 Short-term outcomes

4.1.1 Initial Employment

The first outcome we consider is the likelihood of having observed employment for any duration during the quarter immediately following graduation.[18] Event study coefficients and their 95% confidence intervals are plotted in Figure 3.[19] Because we include detailed major by quarter-of-year fixed effects, all four quarters of 2022 are absorbed as the reference period and plotted as zeros. The red dashed line denotes the release of ChatGPT-3.5 in November of 2022, and the gray shaded box indicates cohorts who graduated after its release. We differentiate between these two time periods to consider whether effects are more or less pronounced among existing participants in the labor market versus new entrants. The distinction is minimal at such a short post-graduation duration, but it becomes more salient when we analyze longer-term outcomes.

In the quarters prior to 2020, the event study coefficients are usually statistically indistinguishable from zero across all deciles, implying that differences in the employment rate across deciles were stable before the pandemic, and that these differences had largely reverted to their long-run levels by 2022. The COVID-19 pandemic and subsequent recovery is not entirely controlled for by the included set of fixed effects. Deciles seven, eight, and ten have negative pandemic coefficients, suggesting that graduates in these majors had worse employment outcomes during the COVID pandemic and relatively muted recoveries compared to graduates from the bottom six deciles. Decile

[18] We refer to this as initial employment, and to the earnings measure as initial earnings. Graduates may have either pre-graduation employment, or employment in the quarter of graduation, but these measures are intended to capture labor market outcomes in the first full period where we can reliably measure them.

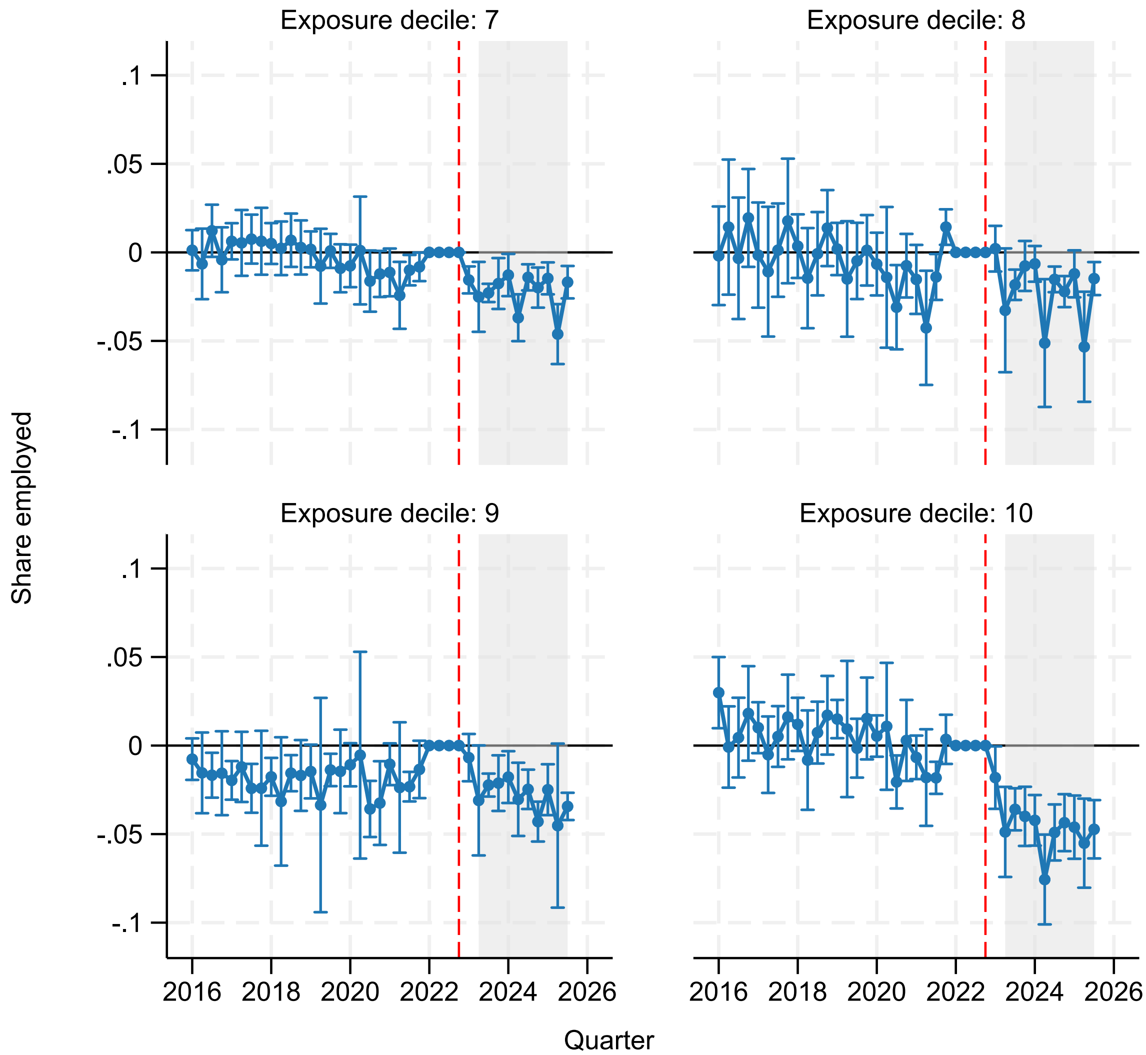
[19] In this figure and all subsequent regression analyses, standard errors are clustered by college major.

nine, by contrast, had a stronger post-pandemic recovery, and many of the pre-ChatGPT event study coefficients for this decile are negative. 2022 may have been an unusually strong labor market for these graduates, and accordingly the subsequent results for this decile should be viewed with this in mind.

Post-ChatGPT, there is a small to moderate decline in employment relative to the control group across the top four deciles with a particularly large effect for the tenth decile. For deciles seven, eight, and especially ten, these are statistically significant declines: an average of two percentage points for deciles seven and eight and five percentage points for decile ten relative to 2022. In these deciles, recent trends also appear to be a distinct departure from the pre-pandemic period. Decile nine also shows evidence of statistically significant declines in employment relative to 2022, and of a similar magnitude to those for deciles seven, eight, and ten; however, since recent coefficient estimates are not statistically lower than the pre-pandemic average, this might reflect reversion to long-run trends rather than a negative effect of AI on this decile.

Overall, the results of Figure 3 suggest that there was an immediate decline in the initial employment rate following ChatGPT's introduction among a sizable share of college graduates, but employment effects are clearly most acute among the most-exposed decile of majors. The relatively flat pre-trends and precise timing help to rule out some alternative explanations, such as heterogeneity in the post-pandemic recovery or the effects of a tight labor market in the reference period.

Figure 3: Share employed one-quarter post-graduation



Notes: Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Estimates from a regression with institution-major, institution-quarter, and major-quarter of year fixed effects. Standard errors are clustered by college major. Error bars denote 95% confidence intervals. Quarter denotes observation period. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 5,685,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

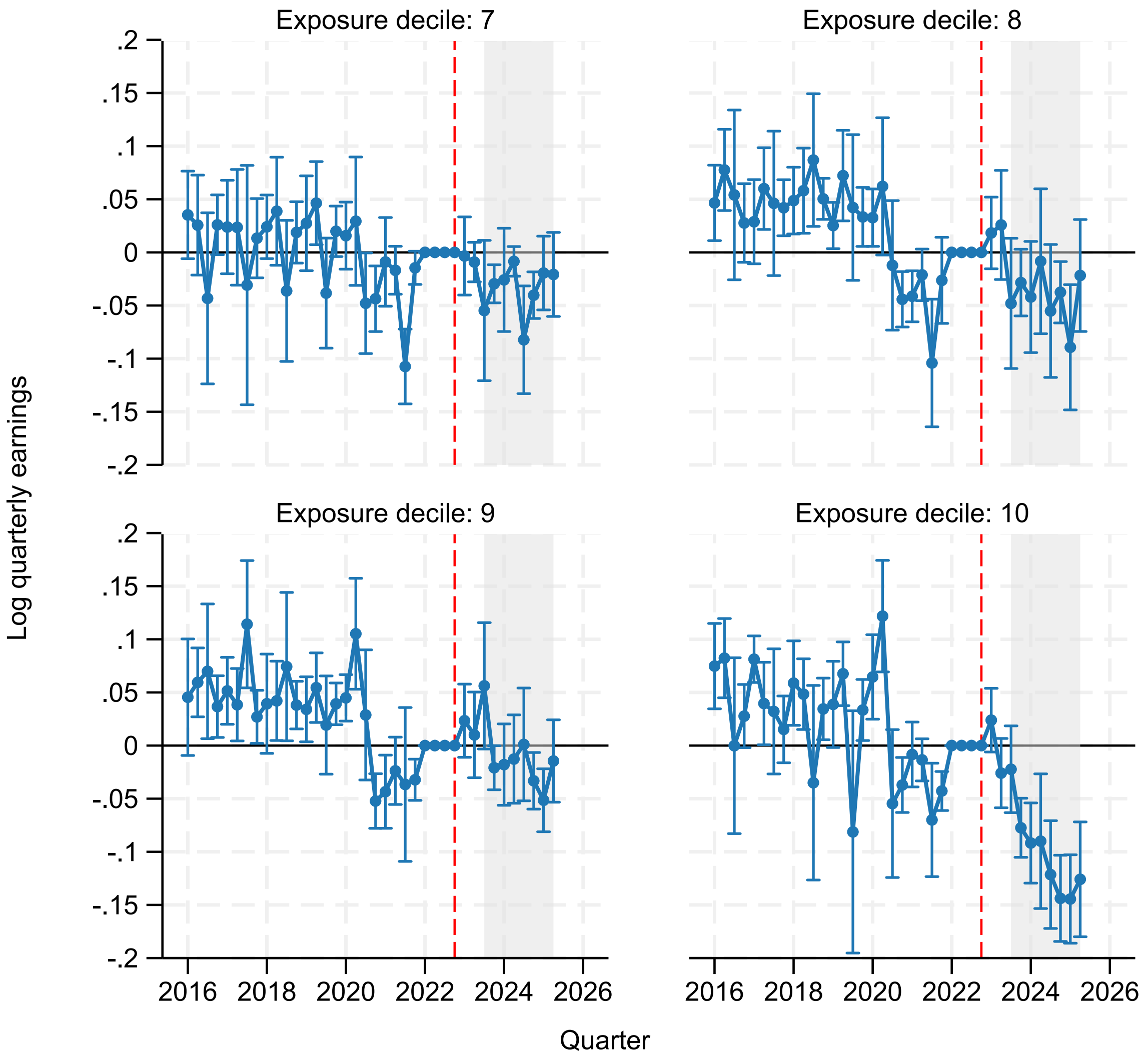
4.1.2 Initial Earnings

Our next outcome, shown in Figure 4, is a measure of average initial earnings, defined as log quarterly earnings in the dominant job for workers with full-quarter employment. To be considered full-quarter employed, a worker must have positive earnings in the previous, current, and subsequent quarter; because of this structure, we measure this outcome two quarters after graduation.

In each plotted exposure decile, the pre-COVID pandemic coefficients are positive and often statistically different from zero: an average coefficient of one log point for decile seven, five log points for deciles eight and nine, and three log points for decile ten. Thus, real earnings in 2022 among more exposed graduates had already begun to decline in comparison to those of less exposed graduates. This could reflect a weaker post-COVID pandemic recovery through 2022 among the occupations and industries that hire these majors. Alternatively, it could reflect a structural trend of declining demand for these graduates' labor that predates ChatGPTs release, such as what might occur due to the rise of remote work. We test this hypothesis explicitly in Section 7.

From 2023 onward, the event study coefficients for deciles seven, eight, and nine are slightly negative but not consistently statistically different from zero. Relative to the long-run average pre-COVID pandemic, they represent meaningful declines: three log points for deciles seven and eight and one log point for decile nine. It is possible that earnings would have recovered to their pre-pandemic average if not for ChatGPT, and the uptick in earnings for students who graduated less than two quarters prior to the release of ChatGPT suggests this might be the case. The top exposure decile, however, has an undeniable decline in earnings following the release of ChatGPT. By 2025, the quarterly earnings for this group were down 14 log points relative to the 2022 change of the least exposed deciles, and even more in comparison to the pre-pandemic average.

Figure 4: Log quarterly earnings two-quarters post-graduation



Notes: Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. Estimates from a regression with institution-major, institution-quarter, and major-quarter of year fixed effects. Standard errors are clustered by college major. Error bars denote 95% confidence intervals. Quarter denotes observation period. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 3,225,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

As we discuss in Section 8, the effects we observe compare in magnitude to the earnings losses of graduates entering the labor market during a recession. The appearance of an incomplete post-COVID-19 recovery among the most AI-exposed graduates motivates our subsequent analysis in

Section 7 to rule out alternative explanations.

4.1.3 Industry

There are two broad explanations for the earnings declines we observe. Either graduates with the most exposed majors are receiving reduced pay to perform the same jobs that prior cohorts were employed in, or they are accepting jobs in a lower-paying range of the employment distribution. This latter possibility is supported by the literature on cyclical effects. Both Altonji et al. (2016) and Oreopoulos et al. (2012) find evidence consistent with such a pattern, either by measuring changes in the prevalence of high-paying occupations, or by constructing measures of matched firm quality. We follow this literature and decompose our estimated earnings effect into the portion due to changes in the industry composition of exposed graduates' initial jobs versus reduced earnings by graduates employed within the same initial industries.

We quantify the shifting industry sector composition of college graduates in Figure 5. The left panel plots the percentage point change in the distribution of graduate employment following graduation by industry sector, defined by NAICS 2-digit sector, from 2022 to 2024 among the bottom six exposure deciles.[20] Positive values reflect a higher share of graduates heading into a sector, while negative values reflect declining shares. Sectors are sorted by their mean 2022 quarterly earnings among this subset of graduates, and data points are scaled by their 2022 employment share among the set of majors. Among these less-exposed graduates, there is a small negative relationship between earnings and sectoral share; the lowest paying industries grew slightly while large industries near the middle of the earnings distribution shrank. The right panel plots the same relationship for the most AI-exposed decile of majors. Here, the pattern becomes pronounced, as the most AI-exposed graduates shifted out of their highest-paying industries and into their lowest paying ones. New graduates in the most-exposed decile became more than two percentage points more likely to end up in Accommodation & Food Services, and similarly more likely to take a job in Retail Trade, despite these being two of the lowest-paying sectors for these graduates. Con-

[20]The employment measure used in this analysis is full-quarter employment two quarters after graduation, aligning it with our measure of initial full-quarter earnings and allowing us to decompose effects.

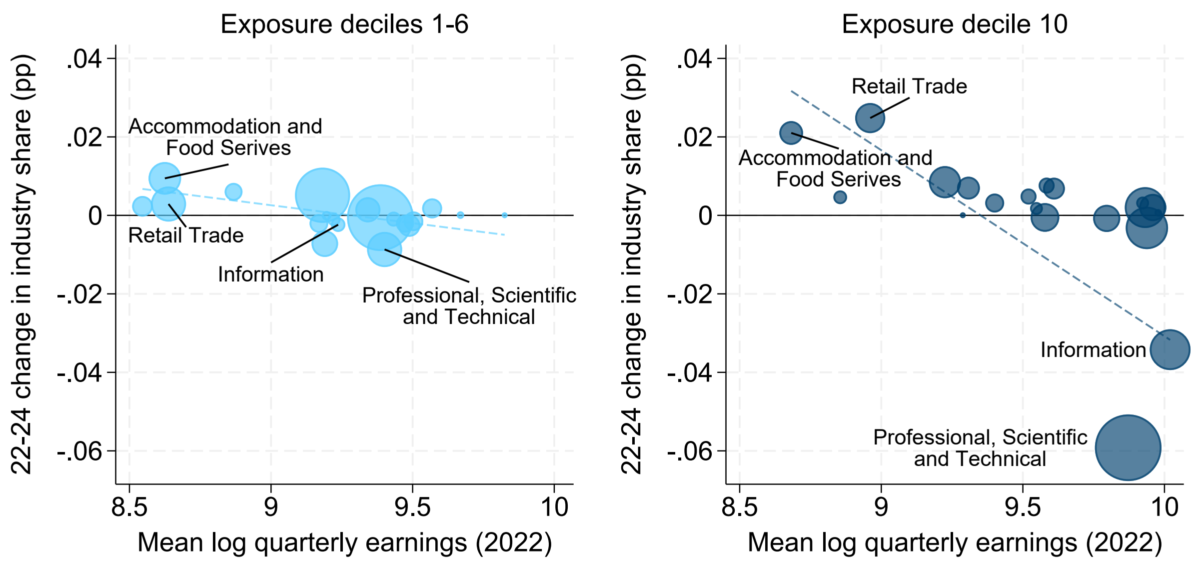
versely, the highest paying sector, Information, shrunk its share of exposed graduate employment by over three percentage points. The share of top-decile jobs in the Professional, Scientific, and Technical sector, the largest source of high paying jobs for highly-exposed majors, contracted by nearly six percentage points. Notably, Tucker (2026) finds that these two sectors have among the highest rates of AI exposure.

Figure 5 highlights the role that industry-level patterns might be playing in outcomes of new graduates. The same adoption costs and scale effects that might lead AI's effects to be nonlinear in occupation- or major-level exposure may also lead its effects to be highly concentrated among specific industries (e.g., Lashkari et al., 2026), and our findings are consistent with the concentrated pattern of industry-level effects found by Tucker (2026). However, sectoral shocks that occur around the time of ChatGPT's introduction could threaten identification of AI's impacts. As we discuss further in Section 7, Tucker (2026) does not find evidence that alternative explanations such as monetary policy shocks are the cause of the observed industry-level patterns. On balance, the evidence supports AI as the primary reason for sectoral declines.

This compositional shift in employment is quantitatively important to the overall earnings changes among exposed graduates that we observe. In Figure 6, we decompose the change in real earnings since 2022 by an "across-industry composition effect", which is the change in industry size relative to 2022 multiplied by mean 2022 earnings, and a "within-industry earnings effect", which is the change in mean quarterly earnings for the industry relative to 2022 multiplied by industry size.[21] When added together, these effects mechanically equal the total change in average quarterly earnings over time. The figure shows that from 2016 to 2022, the vast majority of earnings growth for most majors came from earnings growth within industry, rather than compositional shifts. And, that pattern continues to hold among less-exposed college graduates with respect to earnings declines in more recent years. In contrast, since 2022 approximately half of the overall earnings decline for the top decile of AI-exposed new graduates is explained by their movement into lower-paying sectors. Absent industry shifts, average real earnings in 2024 for AI-exposed

[21] To address seasonality concerns, this figure aggregates average full-quarter earnings based on the year of observation.

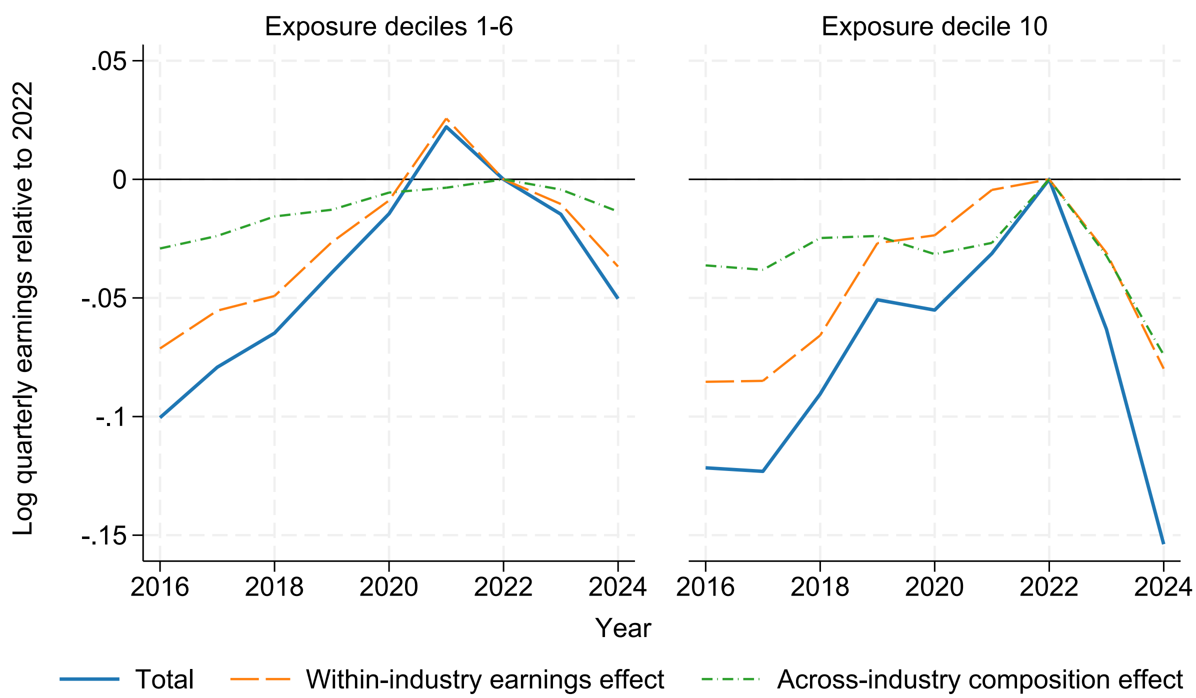
Figure 5: Change in industry composition by AI exposure



Notes: Industry is defined by the 2-digit NAICS code associated with the establishment of a worker's dominant full-quarter job two quarters post-graduation. The change in industry share from 2022 to 2024 is based on graduates from July 2021 to June 2022 (whose two quarter post-graduation period corresponds to January 2022 to December 2022) relative to graduates from July 2023 to June 2024. The mean log quarterly earnings are constructed with the dominant full-quarter earnings of graduates from the specified exposure deciles. A linear trend weighted by exposure-decile industry size is plotted with a dashed line. Number of observations: 512,900. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

majors would still have declined nearly eight percent to below their 2019 levels. However, the additional role of changes in sectoral composition means that total average earnings declined by fifteen percent, placing real earnings at levels last seen before 2016.

Figure 6: Decomposing earnings effect



Notes: Industry is defined by the 2-digit NAICS code associated with the establishment of a worker's dominant full-quarter job two quarters post-graduation. Mean log quarterly earnings are constructed with the dominant full-quarter earnings of graduates from the specified exposure deciles. The within-industry earnings effect is equal to annual industry share multiplied by the change in mean earnings from 2022. The across-industry composition effect is equal to the change in industry share from 2022 multiplied by 2022 earnings. Number of observations: 2,284,900. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

In sum, graduates with majors highly exposed to AI are less likely to be employed in the quarter following graduation than less-exposed graduates. Once they do find employment, they are significantly more likely to work in lower paying industries. Even graduates who find employment in traditionally high paying industries are receiving lower pay on average. Altogether, the short-term effects of AI on the labor market outcomes of highly AI-exposed graduates are markedly negative.

4.2 Longer-term outcomes

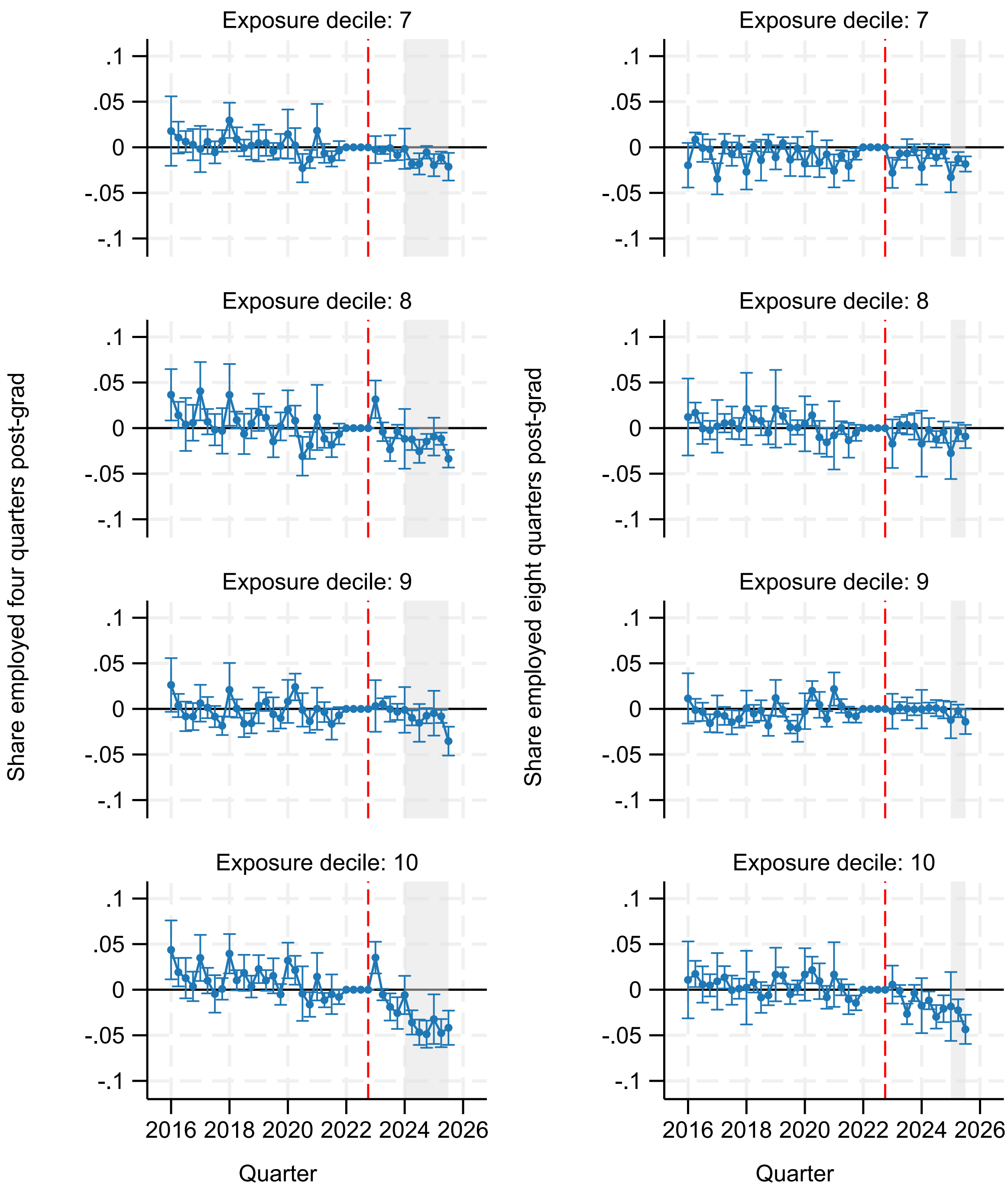
4.2.1 Employment

In the previous section, we documented large, negative declines in employment and earnings for graduates with majors in the top decile of AI exposure. Now, we examine the persistence of those effects. Our window is limited by the relative recency of ChatGPT's release and the lag in availability of the administrative data sources we use. Nonetheless, by looking at outcomes four and eight quarters after graduation, we can begin to see how new graduates' labor market outcomes evolve following the initial post-graduation shock. We can also begin to isolate the role of graduation timing, since the longer time horizons provide additional post-2022 comparisons of cohorts who graduated before ChatGPT's release to those that graduated after.

Figure 7 presents the coefficients from two regressions with the outcomes of employment four quarters post-graduation and employment eight quarters post-graduation. By the end of their first full year post-graduation, graduates with majors in the seventh, eighth, and tenth deciles of exposure have recovered nearly half of their immediate employment declines, but a decline still exists relative to the baseline group: an average of -1.4 percentage points for deciles seven and eight and an average of -4.2 percentage points for decile ten from 2024 onward.

By two years post-graduation, there is minimal evidence of employment declines for graduates with majors in exposure deciles seven through nine relative to majors in deciles one through six. Estimated event study coefficients for the top decile, by contrast, continue to be negative with an average value of -2.0 percentage points from 2024 onward.

Figure 7: Share employed one and two years post-graduation



Notes: Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Estimates from a regression with institution-major, institution-quarter, and major-quarter of year fixed effects. Standard errors are clustered by college major. Error bars denote 95% confidence intervals. Quarter denotes observation period. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 5,747,000 four quarters post-graduation and 5,740,000 eight quarters post-graduation. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

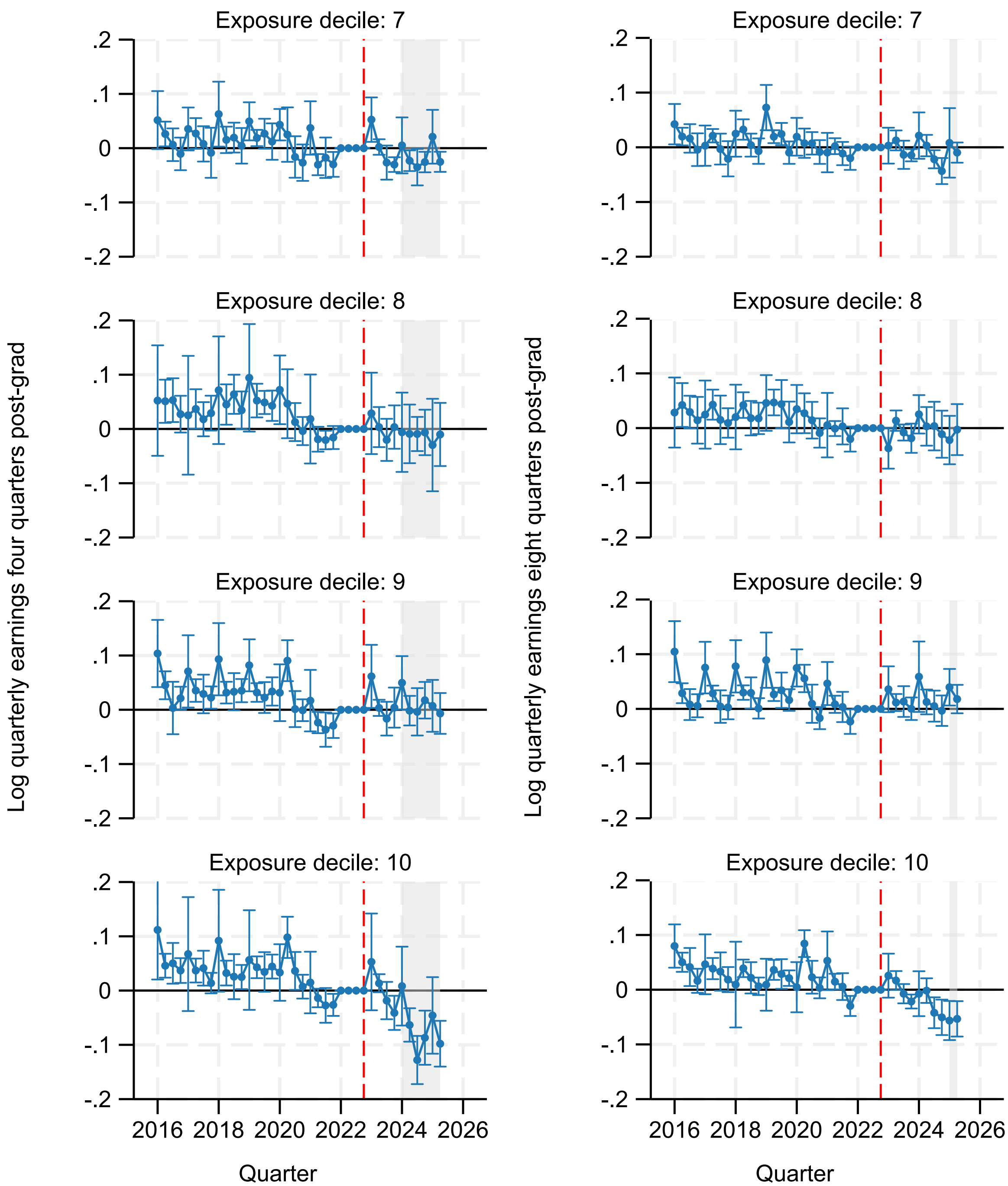
4.2.2 Earnings

When analyzing the period two quarters after graduation in Section 4.1.2, we observed a consistent pattern of statistically significant positive event study coefficients prior to the COVID pandemic, possibly reflecting an incomplete pandemic recovery as of 2022. Yet, when compared against the long-run average, we found evidence of negative effects in exposure deciles seven through nine, and a pronounced decline in average earnings for the most AI-exposed decile. In Figure 8, we evaluate the same outcome four and eight quarters after graduation to see whether earnings patterns converge.

Regardless of exposure decile, the figure shows persistence in the previously observed pattern of positive pre-period coefficients. This is congruent with an underlying pandemic recovery explanation for the observed pre-trends. Moreover, among deciles seven through nine, a consistent story emerges. By one year post-graduation, negative coefficients on earnings among graduates in these deciles attenuate close to zero. Despite the small declines in employment rate observed at one year in the previous subsection, this suggests that graduates in these majors ultimately find jobs that pay a similar wage premium to the ones they might have expected to obtain before ChatGPT's release, and that they do so within one to two years.

In the most AI-exposed decile, the story is substantially less reassuring. Statistically significant negative effects on average full-quarter earnings persist for at least two years among post-2022 graduates. One year out of school, earnings have declined by an average of 8.1 log points from 2024 onward, and two years post-graduation, earnings remain 3.3 log points lower than they would have been if they had followed the trend of the six least-exposed deciles.

Figure 8: Log quarterly earnings one and two years post-graduation



Notes: Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. Estimates from a regression with institution-major, institution-quarter, and major-quarter of year fixed effects. Standard errors are clustered by college major. Error bars denote 95% confidence intervals. Quarter denotes observation period. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 3,666,000 four quarters post-graduation and 3,814,000 eight quarters post-graduation. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

Overall, these longer-term earnings and employment results suggest that the impacts of AI on most graduates are reasonably transitory. While AI may be having an impact on initial job finding and subsequent earnings, these effects do not persist for long among graduates of majors in exposure deciles seven through nine. Although the available time series is short and AI's capabilities and impacts may evolve, there is little evidence here to suggest long-term scarring among such graduates. In contrast, the results paint a concerning picture of the labor market for graduates in the most AI-exposed decile of majors, including the possibility of persistent effects to earnings over several years.

4.2.3 Job switching

Our final baseline outcome of interest is job switching, shown in Figure 9. As previously defined in Section 2.4, we flag a job switch as occurring when a graduate's observed dominant job was with a different firm than the firm of their dominant job eight quarters post-graduation. We also showed that job switching of recent graduates reached a peak in 2022, frequently attributed to the unwinding of the pandemic and the unusually tight labor market at that time. This complicates the choice of a reference period, and for this reason, we find it more instructive to compare post-2022 coefficients to the long-run pre-2022 average, rather than to 2022 alone. While the magnitude of coefficient point estimates reflects the difference from 2022, differences in coefficients between other time periods can be interpreted as the percentage point decline in the probability of a job switch between those periods.

Figure 9 shows that among the top four deciles of exposure, job switching probabilities fell from their 2022 peak back to COVID-pandemic levels through much of 2023 and 2024 (average coefficient of -5.9 percentage points in 2024) before increasing rapidly in 2025 through the end of our data series (average coefficient of -2.7 percentage points in 2025). The timing of higher job switching corresponds precisely with the cohorts of students who graduated after the release of ChatGPT. Students who graduated prior to ChatGPT but reached their two-year tenure in the labor market post-college after the release of ChatGPT were less likely to switch jobs, similar to students who were two years removed from college during the COVID pandemic. This suggests that pre-ChatGPT graduates who already found jobs before ChatGPT's release may have foregone voluntary job switching in light of a weak labor market. Conversely, post-ChatGPT graduates were comparatively more likely to switch, potentially because the weak labor market at time of entry led them to take lower-paying or less desirable initial jobs that they later sought to leave. Although this timing pattern is observed across exposure deciles, the eight-quarter earnings figures of the preceding subsection suggest that graduates of the most-exposed decile may have been less successful in switching to higher-paying jobs.

5 Results by major

Thus far, we have presented the event study coefficients from Eq. 1 with college majors grouped into deciles. Discretizing exposure is useful for concisely identifying non-linearities in the effects of AI, as well as pooling together small majors or rare double-major combinations to assist in statistical inference, but no part of our empirical strategy strictly requires us to do so. We additionally estimate event studies with the same fixed effect structure but major-specific year coefficients rather than decile group. This provides point estimates of effects for all sufficiently popular majors and double-major combinations.[22]

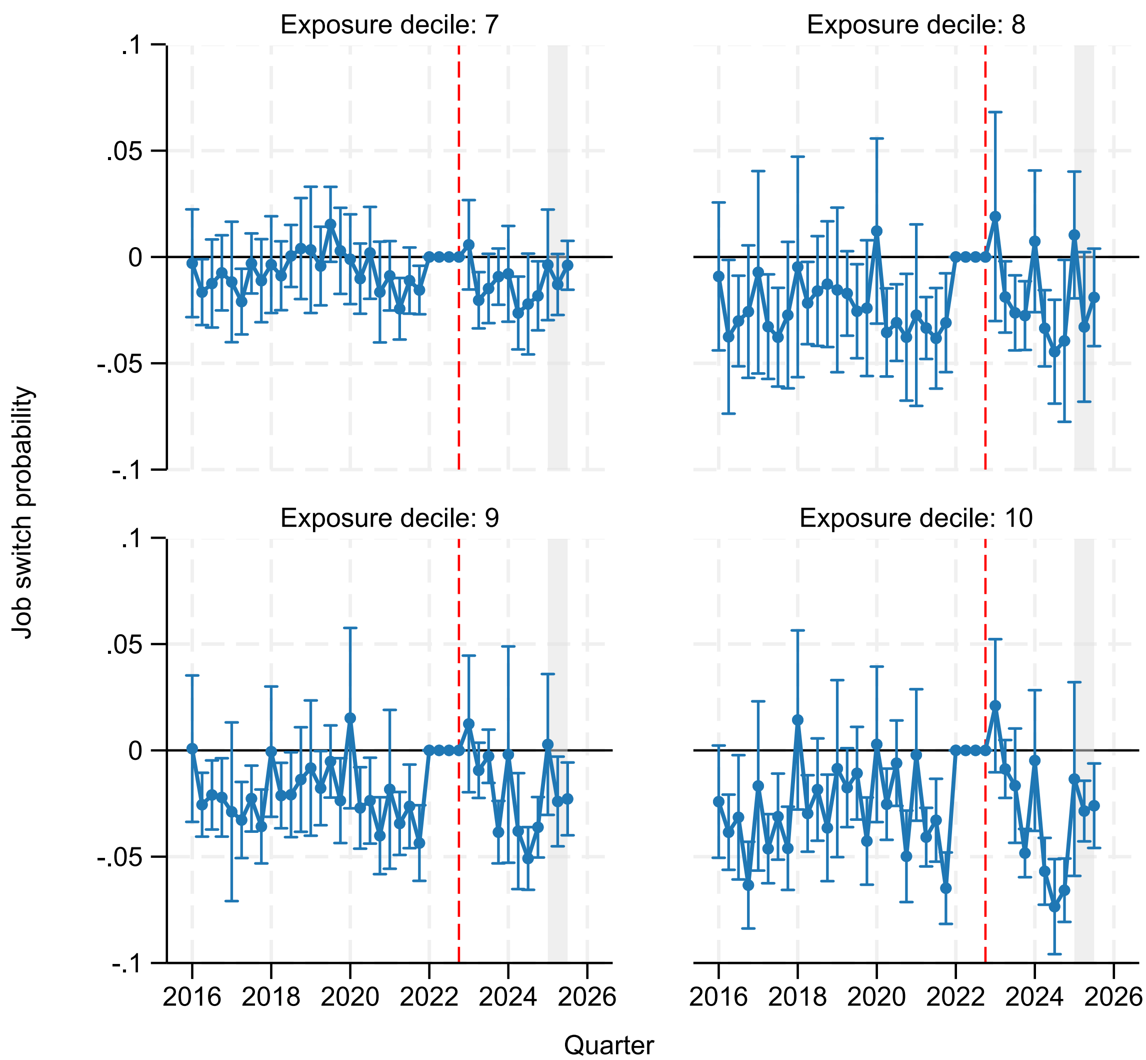
Our major-specific estimating equation is:

$$y_{it} = \sum_{s \neq 2022} \sum_{m} \beta_{ms} 1[m(i) = m, t = s] + \delta_{c(i),m(i)} + \phi_{c(i),t} + \tau_{m(i),q(t)} + \varepsilon_{it}. \qquad (2)$$

Each fixed effect term has the same definition and interpretation as in Equation 1; however, the coefficients β_{ms} now denote the difference between major m's outcome in year s and its corresponding outcome in 2022. Note that for the purposes of regression, an arbitrary major was chosen as the reference major in each year; we recenter the coefficients to be relative to the overall aver-

[22]Majors or double-major combinations with fewer than 1,000 observations across all years are grouped together in a residual category. This residual group is not plotted in subsequent figures, but this ensures that all graduates continue to contribute to identification of fixed effects.

Figure 9: Job switching eight-quarters post-graduation



Notes: Job switches are identified by a change in the State Employer Identification Number (SEIN) of a graduate's dominant employer eight quarters post-graduation relative to the SEIN of the first dominant job held post-graduation. The denominator is all graduates with positive earnings in the eighth quarter post-graduation. Estimates from a regression with institution-major, institution-quarter, and major-quarter of year fixed effects. Standard errors are clustered by college major. Error bars denote 95% confidence intervals. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 4,660,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

age across all majors in the corresponding year, so that plotted point estimates reflect the change in employment or earnings since 2022 relative to the weighted average change. When comparing with the prior decile-based results, one should keep in mind this change in reference group and interpretation.

Figures 10 and 11 present scatterplots of the major-specific outcome coefficients for the year 2024. Each major is represented by a bubble scaled to the number of graduates from 2018-2022, and its position on the horizontal axis corresponds to their standardized baseline GPT-4 Beta exposure score. The decile thresholds used to previously classify majors are denoted by alternating gray bars. Selected majors are labeled, including ones at the extremes of the exposure distribution. Appendix Table A3 contains the number of graduates for each major, their exposure score, and the 2024 event study coefficients and standard errors for reference.

As Figure 10 shows, major-level initial employment outcomes are clearly negatively correlated with AI exposure. The vast majority of graduates in the bottom two deciles have higher employment rates in 2024 versus 2022, and graduates in deciles three through six are split between majors with positive and negative employment changes. An inflection occurs at decile seven, and nearly all majors in this decile or higher have negative employment changes. Decile ten, which is predominantly composed of computer science and adjacent majors, has the most negative employment outcomes, mirroring our previous baseline results.

Figure 11 shows results for initial earnings. Consistent with our baseline regression results, the figure shows that the relationship between earnings and exposure is more nonlinear than it is between employment and exposure, and more dramatic within the top decile. The majors in the lowest decile of exposure universally had 2022-2024 earnings growth above the overall average, and almost all the majors in the tenth decile of exposure have substantially lower earnings growth, but the majors in-between are nearly uniformly distributed between positive and negative earnings. As discussed in Section 4.1.2, this is partially due to the use of 2022 as a reference period; new graduate initial earnings in 2022 were already below pre-pandemic levels, possibly because of an incomplete post-pandemic recovery. Comparing 2024 to the longer run average would likely yield a steeper slope over the exposure distribution.

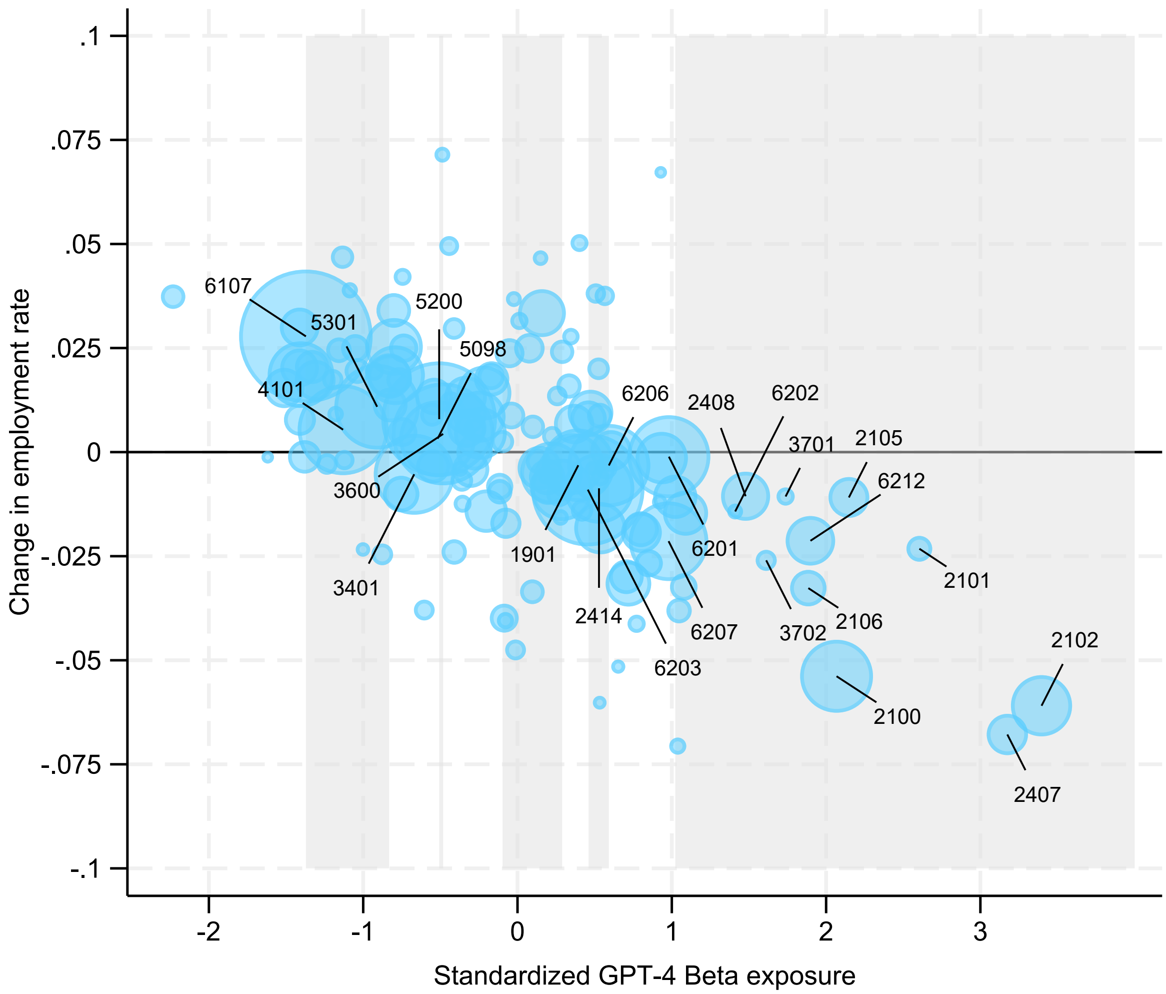
A couple of very small fields notwithstanding, the three majors with the largest earnings declines – Computer Science, Computer and Information Systems-General, and Computer Engineering – are also the three majors with the largest employment declines. Fewer of these graduates found jobs immediately after graduation, and those who did so received lower earnings. Also, recall that our measure of initial earnings is dominant full-quarter earnings, so the earnings effects are unlikely to be a result of delayed employment reducing the numbers of weeks worked in the measurement period.

6 Heterogeneity for highly exposed occupations

Our baseline measure of AI exposure at the college major level reflects the average *ex ante* exposure of a graduate entering the labor market with that major. One concern with this mean-based mapping of majors to occupational exposure is that it might mask important variation across majors who have the same mean exposure but different underlying distributions of that exposure. The potential for negative outcomes might correspond not just to how exposed a graduate expects to be on average, but to how likely they are to obtain a highly AI-exposed job. As discussed in Section 2.2, firms' labor substitution decisions are a function of comparative advantage, adoption costs, and task scale, all of which might lead firms to invest in labor substitution only at the highest levels of exposure. Moreover, if workers' willingness to take a lower-paying but less exposed job in another occupation (similar to the pattern we observe for industries) depends on nonlinear future displacement risk, then the contribution of these compositional effects to overall earnings losses will similarly depend on the extent of high exposure rather than the average level of exposure across all jobs.

Consider, as a stylized example, two majors with nearly identical average AI exposure across occupations: Journalism and Accounting. Suppose that Journalism majors work exclusively in two occupations: Broadcast Journalist, which has a low AI exposure score, and Proofreader, which

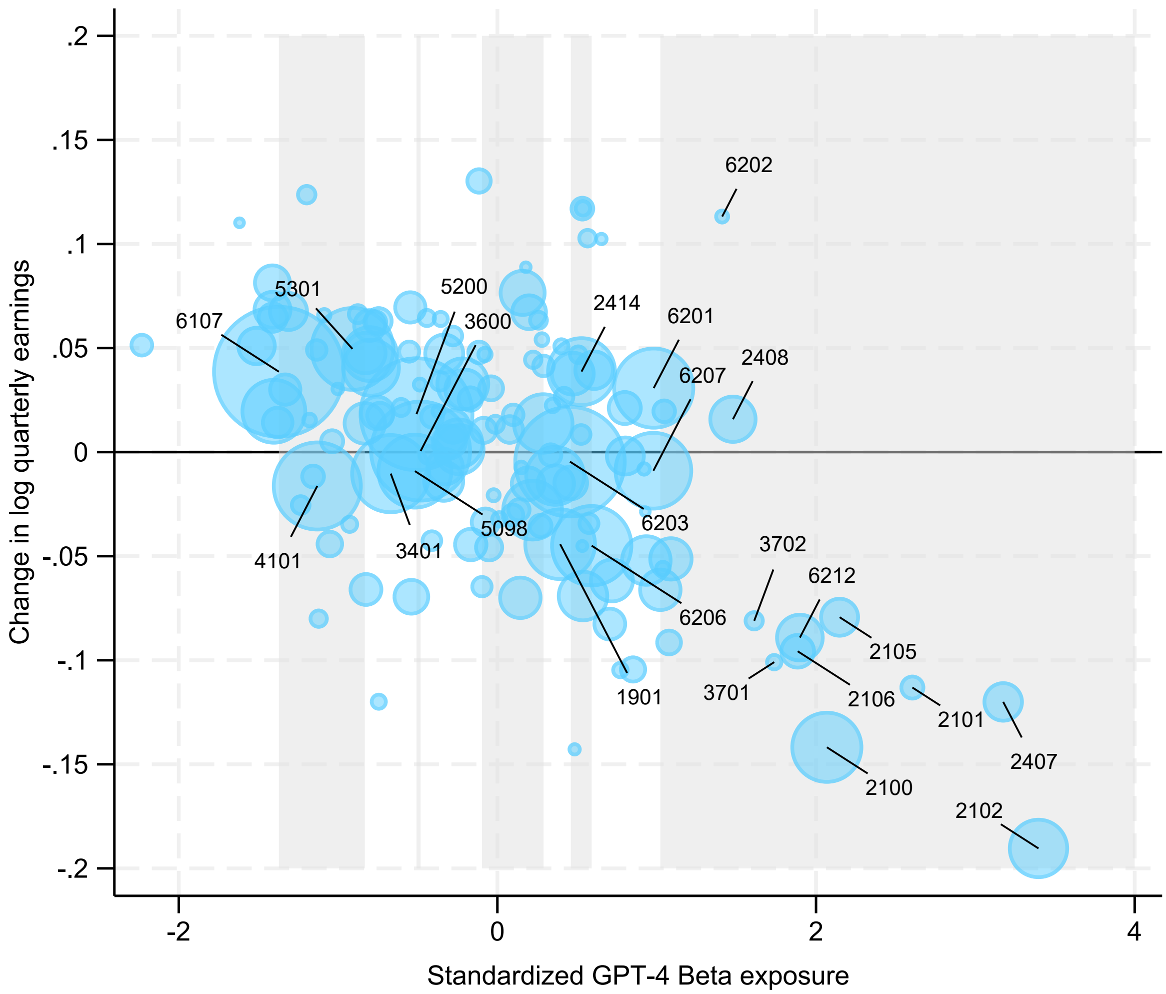
Figure 10: Initial employment by major



1901: Communications
2100: Computer and Info. Systems-General
2101: Computer Programming
2102: Computer Science
2105: Info. Sciences
2106: Computer Admin. Mgmt. and Security
2407: Computer Engineering
2408: Electrical Engineering
2414: Mechanical Engineering
3401: Liberal Arts
3600: Biology
3701: Applied Mathematics
3702: Statistics
4101: Physical Fitness, Parks, Rec., and Leisure
5098: Multi-disciplinary or General Science
5200: Psychology
5301: Criminal Justice and Fire Protection
6107: Nursing
6201: Accounting
6202: Actuarial Science
6203: Business Mgmt. and Admin.
6206: Marketing
6207: Finance
6212: Mgmt. Info. Systems and Statistics

Notes: Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Estimates based on a regression of employment on major-year coefficients with institution-major, institution-quarter, and major-quarter of year fixed effects. Reference year 2022. Coefficients re-centered relative to overall year average. Bubble sizes correspond to count of graduates from 2018 to 2022. Only coefficients for the year 2024 and majors with at least 1,000 graduates between 2018 and 2022 are plotted. Shaded regions denote alternating deciles of exposure. Number of observations: 5,685,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

Figure 11: Initial earnings by major



1901: Communications
2100: Computer and Info. Systems-General
2101: Computer Programming
2102: Computer Science
2105: Info. Sciences
2106: Computer Admin. Mgmt. and Security
2407: Computer Engineering
2408: Electrical Engineering
2414: Mechanical Engineering
3401: Liberal Arts
3600: Biology
3701: Applied Mathematics
3702: Statistics
4101: Physical Fitness, Parks, Rec., and Leisure
5098: Multi-disciplinary or General Science
5200: Psychology
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6107: Nursing
6201: Accounting
6202: Actuarial Science
6203: Business Mgmt. and Admin.
6206: Marketing
6207: Finance
6212: Mgmt. Info. Systems and Statistics

Notes: Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. Estimates based on a regression of earnings on major-year coefficients with institution-major, institution-quarter, and major-quarter of year fixed effects. Reference year 2022. Coefficients re-centered relative to overall year average. Bubble sizes correspond to count of graduates from 2018 to 2022. Only coefficients for the year 2024 and majors with at least 1,000 graduates between 2018 and 2022 are plotted. Shaded regions denote alternating deciles of exposure. Number of observations: 3,225,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

has a high exposure score and might hypothetically be substituted with AI. By contrast, suppose Accounting majors work exclusively in occupations with moderately high exposure, below the level where firms might opt to substitute. If negative employment and earnings outcomes are driven by current labor substitution, then we would expect Journalism majors to be more negatively impacted than Accounting majors.

To test this possibility, we construct an alternative major-level exposure measure based on the share of the major's employment in the top occupational AI exposure quintile. The distribution of this measure is compared with mean occupation scores in Appendix Figure A1. Note that every major has some share of their occupations in the top quintile of exposure, and 82% of majors have at least ten percent of their occupations in the top quintile. Even the majors we classify in our baseline reference group could experience significant reductions in their labor market opportunities if effects are driven by only the most exposed occupations being replaced with AI.[23] In keeping with our previous stylized example, Journalism majors have 36% of their occupational distribution in the top exposure quintile while Accounting majors have only 16% of their occupation distribution in the top exposure quintile, despite their nearly identical mean exposure.

In general, the two exposure measures are highly correlated, but the correlation is highest at the bottom (correlation=0.78 in baseline deciles 1 through 6) and very top of the distribution (correlation=0.87 in decile ten). More variation in the share-based measure is found among baseline deciles seven, eight, and nine (correlation=0.29, 0.21, and 0.16, respectively). It is this variation that permits us to isolate the potentially distinct relationship between these two measures of exposure and our outcomes of interest.

Figure 12 plots each major in the seventh through ninth deciles of mean occupation score along with a trend line between the major's share of occupations in the top exposure quintile and the major's 2024 event study coefficient in the initial employment and earnings regressions

[23]This finding may also reflect that many college majors obtain many general skills, and that such skills can be applied to a broad range of occupations (e.g., Gathmann and Schönberg, 2010). Prior research has found that future earnings are positively associated with the occupational specificity of one's college major, and that graduates who work in fields unrelated to their field of study earn less (Kinsler and Pavan, 2015; Leighton and Speer, 2020). To the extent that AI might be negatively impacting all new graduates, our methodology based on comparisons across exposure deciles may understate the magnitude of effects.

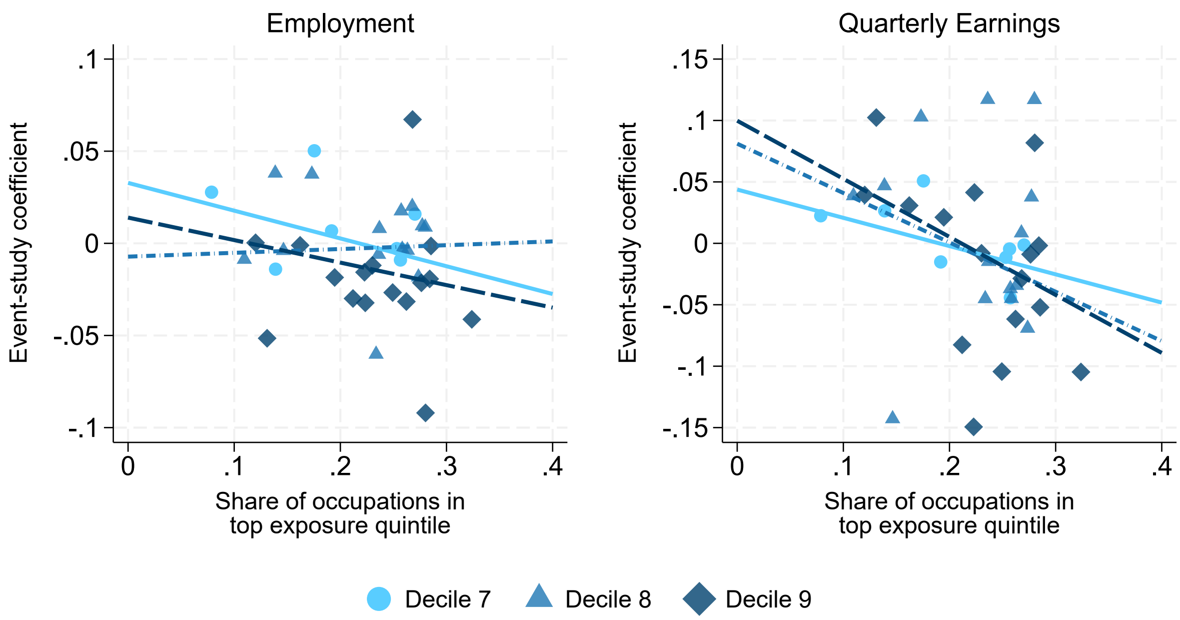
from Figures 10 and 11. Due to the high correlation between mean occupation score and share of occupations in the top quintile, we exclude the other deciles. The left panel shows that a ten percentage point increase in the share of occupations in the top exposure quintile is associated with a 1.5 percentage point lower initial employment share for decile seven and a 1.2 percentage point lower employment share for decile nine.[24] These are substantial magnitudes in comparison to the mean 2024 event study coefficients for the majors in these deciles (-0.005 and -0.013, respectively). The relationship between the share of occupations in the top exposure quintile and employment for the eighth decile is effectively zero. The right panel shows an even stronger relationship between a major's share of occupations in the top exposure quintile and its decline in initial earnings. A ten percentage point increase in the share of occupations in the top exposure quintile for the seventh, eighth, and ninth deciles is associated with decreases of 2.3, 4.0, and 4.8 log points of quarterly earnings, respectively. The mean event study coefficients for these deciles in 2024 are -1.3, -0.7, and -0.8 log points. Thus, variation in the share of occupations in the top quintile is highly informative of the earnings effects by major within the interior exposed deciles.

We also re-estimate our baseline event study with the share of each major's occupations in the top quintile of exposure treated as a continuous explanatory variable with time-varying effects. The results are in Appendix Figure A2. Across all majors, a ten percentage point increase in the share of a major's occupation in the top exposure quintile is associated with a 1.3 percentage point decline in the likelihood of employment one quarter post-graduation and a 3 log point decline in initial quarterly earnings from 2024 onward relative to 2022.[25]

[24]The standard deviation in the share of occupations in the top quintile is 0.12, so this is equivalent to an increase of roughly 83% of a standard deviation.

[25]Callaway et al. (Forthcoming) show that a two-way fixed effects model with continuous treatment, such as the one estimated here, cannot recover the true global average causal response, but instead recovers a weakly causal weighted average of treatment effects under a stronger parallel trends assumption. Any interpretation of these coefficients should be done with caution. Nonetheless, after applying these coefficients to the occupation share averages for each decile, these regressions predict a 4.3 percentage point decline in employment and a 10.9 log point decline in initial earnings for the most-exposed decile from 2024 onward, slightly smaller than the baseline estimates of 4.8 percentage points and 12.6 log points shown in Figures 3 and 4, respectively.

Figure 12: Initial labor market outcomes and share of occupations in top quintile



Notes: Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. Estimates based on a regression of employment or earnings on major-year coefficients with institution-major, institution-quarter, and major-quarter of year fixed effects. Reference year 2022. Coefficients re-centered relative to overall year average. Only coefficients for the year 2024 and majors with at least 1,000 graduates between 2018 and 2022 are plotted. A linear trend weighted by number of observations in each major is also plotted. Number of observations: 5,685,000 in employment regression and 3,225,000 in earnings regression. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

7 Alternative explanations

7.1 Work-from-home

Our fixed effects specification identifies the effects of AI conditional on the assumption that there are no time-varying major-specific shocks correlated with AI exposure. If such a shock does exist and coincides with ChatGPT's release, it might lead one to misattribute changes in outcomes to AI. In a recent paper, Lambert and Schindler (2026) argue that this is the case with work-from-home. The authors find that almost all of the labor market declines that other authors attribute to AI are likely the result of increased work-from-home rates following the COVID pandemic. The challenge in disentangling the two explanations arises from their high correlation. At the occupation level, Lambert and Schindler document a 0.77 correlation between work-from-home and AI exposure; we find that the correlation rises to 0.88 at the college major-level. In general, high underlying correlations such as these can make regression estimates noisy and unstable, limiting the ability to distinguish between explanatory covariates.

We take two approaches to incorporate work-from-home into our estimates, the details of which are provided in Appendix C. Although neither approach fully resolves the aforementioned collinearity issue, they represent the natural extension of existing approaches in the literature to our setting. First, we augment our baseline model by controlling for a continuous work-from-home rate estimated for each college major and year using the ACS PUMS.[26] The revised coefficients provide little evidence to suggest that our estimates are being driven by the rise of remote work. Our second specification, which is a replication of Lambert and Schindler (2026), fixes major-level work-from-home rates at their 2022 values, and both AI exposure and work-from-home are standardized and treated as continuous variables with separate event study series. This model clearly suffers from collinearity, and the pre-period event-study coefficients move in opposite directions

[26]We construct work-from-home rates as the fraction of employed bachelor's degree holders aged 21-29 that answered the question "how did [you] usually get to work last week" with "worked from home" in the ACS. Responses are aggregated by field of degree and survey year and weighted by ACS person weights. Alternative measures involve the share of new job positions that list work-from-home (Hansen et al., 2023) or assessing the feasibility of remote work for each occupation (Dingel and Neiman, 2020).

away from zero when neither AI nor work-from-home should be affecting the labor market in this time period. The post-period coefficients for initial employment are nearly identical to those of our baseline model, and the coefficients for initial earnings attenuate up toward zero by the same amount the pre-period coefficients also increased, so the difference relative to the long-run average is preserved. In sum, although our ability to isolate the effects of work-from-home and AI exposure is limited, available evidence indicates that our results are robust to this concern.

7.2 Oversupply of exposed graduates

The number of graduates with majors at the very top of the exposure distribution grew dramatically in recent years. Figure A3 shows that total graduates in top decile majors increased 24% from academic year 2017 to 2022, presumably reflecting strong interest in the computer-related subjects that compose this decile. Conversely, the number of graduates with majors in the bottom six deciles changed little. This suggests that the employment and earnings declines we observe could simply be a result of an oversupply of graduates with AI-exposed majors relative to a constant or slower-growing demand for their labor.

Overall, trends in new graduate absorption provide limited evidence for this hypothesis. The figure shows that the rise in the number of top-decile graduates was met with a nearly identical rate of increase in the number of initially employed graduates, except for a brief period due to COVID-19. This suggests that demand for these majors was sufficient to absorb supply and hold the employment rate constant through 2022. From academic year 2022 to 2024, the growth of the most exposed majors slowed to under 3% per year, but the count of employed graduates broke trend and declined 3% over the period. No other exposure decile experienced divergence in the number of graduates and the level of initial post-graduation employment.

7.3 Alternative economic activities: Self-employment and graduate school

The LEHD jobs data cover almost all wage earners in the United States. Nonetheless, new graduates could be engaged in other productive economic activities that would not be measured as employment in these data. Most notably, LEHD data do not include information on self-employment, nor do they contain information on graduate school enrollment, both of which often occur in lieu of covered employment. To the extent that rates of self-employment or graduate school enrollment changed differentially across exposure deciles from 2022 onward, we would classify these as a decrease in employment. Alternatively, if these activities are coupled with covered wage and salary employment (such as a part-time job while enrolled), we might observe larger declines in average earnings than would be observed absent the change in the composition of economic activities. To assess these possibilities, we evaluate both outcomes using responses from 21-24 year old bachelor degree recipients in the ACS.[27]

The results of this analysis are shown in Appendix Figures A14 and A15. From 2022 to 2024, the self-employment rate among graduates in the top decile of AI exposure increased by 0.48 percentage points, from 1.01 percent to 1.49 percent, while the self-employment rate for the bottom six deciles of graduates increased by only 0.06 percentage points. Over the same time period, the graduate school enrollment rate among graduates in the top decile increased by 1.66 percentage points, from 16.20 percent to 17.86 percent, while the graduate school enrollment rate for the bottom six deciles decreased by 0.50 percentage points.[28] Taken at face value, changes in self-employment and graduate school enrollment may explain up to half of the five percentage point decline in initial employment of the most exposed graduates that we observe in our baseline estimates.

While this analysis suggests the role of these alternative activities is substantial, it need not imply that our baseline estimates overstate AI's effects, in particular on earnings. For example, workers might be opting into self-employment or graduate school precisely because they cannot

[27]There are a few caveats to this analysis. First, the ACS does not contain date of graduation, so analysis using the ACS may not capture outcomes immediately following graduation. Second, the ACS data may not be comparable to our primary analytic sample, for reasons including the age restriction we apply to the ACS, the institutional composition of our graduate data, and the inclusion in the ACS of individuals who obtained their undergraduate degree abroad, which is a large fraction of graduate school enrollees in fields like Computer and Information Sciences (National Center for Science and Engineering Statistics, 2026). Finally, we are unable to control for time-varying institution effects in the ACS, meaning that some changes in institutional composition and local labor market effects may impact these estimates.

[28]This pattern also largely aligns with comprehensive data from U.S. academic institutions, which shows that graduate enrollment in Computer and Information Science graduate enrollment increased rapidly from 2020 to 2022, peaked in 2023, and then modestly declined in 2024 (National Center for Science and Engineering Statistics, 2026). Note that ACS data do not distinguish field of graduate study, so our estimates include changes to enrollment in other fields that might occur in response to labor market conditions.

obtain traditional employment. Conversely, workers could opt into self-employment to take advantage of new opportunities brought about by the new technology, though this appears unlikely to be the case, at least in terms of current income. Appendix Figure A16 compares median earnings of self-employed and traditionally-employed graduates aged 21-24 in the ACS across deciles of major-level AI exposure.[29] It shows that self-employed recent college graduates earn less than their traditionally-employed colleagues, in particular among the most AI-exposed college majors. The self-employed earnings gap narrows from 2022-2024, but it remains large in the top decile where young self-employed graduates earn less than a third of their traditionally-employed counterparts. This suggests that the increase in self-employment could be in part attributable to weak labor demand; however, the overall effect of self-employment on our estimates of earnings effects depends on where these self-employed individuals would sit in the earnings distribution if they held a wage and salary job instead.

7.4 Other factors

Monetary tightening provides another possible explanation. Interest rates rose sharply during 2022, and their lagged effects could coincide with the diffusion of ChatGPT, particularly because computer-related majors comprise a large share of the top exposure decile, and these graduates are particularly likely to enter into specific industries. Tucker (2026) finds that AI-exposed industries are not substantially more sensitive to monetary policy shocks on average. A historical decomposition suggests that monetary policy shocks through 2023 can account for at most about one-quarter of the relative decline in early-career employment through 2025Q2, and these shocks do not reproduce the observed sharp decline in early-career hiring around the introduction of ChatGPT. Monetary tightening may therefore have contributed to the broader weakening of the early-career labor market, but it cannot account for the full differential change among AI-exposed workers.

Other contemporaneous changes, including the post-pandemic normalization of hiring, the contraction in technology-sector employment, changes in college enrollment and major choice, and

[29] Since the ACS asks about earnings over the last 12 months, we restrict to individuals who worked at least 40 weeks over that period. This likely excludes the most recent graduates but preserves comparability across groups.

shifts in demand for particular skills, are more difficult to separate from the effects of generative AI. Our fixed effects absorb shocks common to graduates from the same college and persistent differences across college-major programs, but they do not account for time-varying shocks to particular fields that coincide with the diffusion of generative AI. The movement of exposed graduates away from their traditional industries, together with evidence that hiring declines associated with AI exposure occur within a broad set of sectors (Tucker, 2026), makes a narrow technology-sector explanation less compelling. Considered jointly, the sharp timing of the estimated changes, their concentration at the top of the exposure distribution, the accompanying sectoral reallocation, and their robustness to the alternative explanations mentioned here point to a substantial role for generative AI in shaping the labor market entry of recent graduates. These findings do not rule out every contemporaneous shock correlated with AI exposure, but they are suggestive that the technology has already begun to alter the opportunities available to graduates from the most exposed fields.

8 Discussion

Overall, our results suggest that there was a fairly immediate and concentrated deterioration in the early labor market outcomes of graduates from the most AI-exposed fields following the introduction of ChatGPT-3.5. Our results point to a disruption that operates most strongly through labor market entry and the quality of the initial jobs obtained by graduates from the most AI-exposed fields. Following the introduction of ChatGPT, graduates from these fields became less likely to find employment immediately after graduation and experienced substantial earnings declines. Among those who became employed, an important part of the adjustment occurred through changes in the sectors they worked. Employment shifted away from relatively high-paying sectors such as Information and Professional, Scientific, and Technical Services and toward lower-paying sectors such as Retail Trade and Accommodation and Food Services. The differential effects become smaller as workers move further from graduation, and job-switching rates increase among the cohorts that entered the labor market after ChatGPT. These patterns are consistent with some exposed graduates initially accepting weaker matches and subsequently moving toward better jobs, although employment and earnings for the most exposed graduates do not fully recover within our measurement window.

These effects help reconcile seemingly different findings in the emerging literature. Studies of hiring and employment in AI-exposed occupations and industries find sizable declines among young workers (Brynjolfsson et al., 2025; Tucker, 2026), while studies following workers already employed in exposed occupations find little evidence of increased unemployment (Eckhardt and Goldschlag, 2025; Massenkoff and McCrory, 2026).[30] Because our exposure measure is fixed by college major before these outcomes are realized, we can follow this adjustment without conditioning on the occupation or industry in which a graduate is ultimately observed. Graduates from highly exposed majors remain strongly attached to the labor market and their adjustment occurs importantly at labor market entry. They are less likely to enter employment immediately and are more likely to sort into lower-paying sectors when they do. This, however, comes at a substantially larger earnings effect than the changes observed within the employed individuals in an exposed sector. Thus, modest employment or unemployment effects can coexist with substantial deterioration in job quality and early-career opportunities.

The pattern we observe closely resembles the effects of entering the labor market during a recession. Weak demand at graduation reduces employment and earnings and shifts new graduates toward lower-paying occupations, employers, and industries (Oreopoulos et al., 2012; Altonji et al., 2016). Our estimated decline in quarterly earnings of approximately thirteen percent shortly after graduation is larger than the nine to ten percent initial annual earnings losses estimated in the recession literature.[31] An important difference, however, is that recessions generate broad aggregate labor market slack, while the shock we study is concentrated among a subset of fields. Less-exposed parts of the labor market may therefore absorb affected graduates, helping to explain

[30]Classifying unemployed workers by exposure is also difficult because prior occupation and industry are not consistently reported in survey data (e.g., Gimbel et al., 2026).

[31]Our measure attenuates to about seven percent after one year, but our earnings measures condition on full-quarter employment and do not include partial-period employment. Much of the recession literature, by contrast, measure annual earnings. Annual earnings additionally capture losses from delayed or intermittent employment, making the magnitudes not directly comparable. Because of this, we consider our estimates to be consistent with the magnitude of a recession.

the modest employment decline alongside substantial movement into lower-paying industries.

Several mechanisms could generate or interact with the labor market effects we observe, although our data do not allow us to fully distinguish among them. The most direct is substitution of tasks previously performed by new graduates. AI may also transform jobs without eliminating them. When some tasks are automated, workers may specialize in the tasks that remain, and the effect on labor demand will depend on workers' comparative advantage across the full task bundle as well as user costs of AI (Freund and Mann, 2026; Lindenlaub et al., 2026). These mechanisms provide one explanation for why our results are concentrated among the most exposed majors. AI-driven changes in demand may also amplify these effects at labor market entry. Huckfeldt (2022) shows that hiring becomes more selective during downturns and pushes displaced workers into lower-skill jobs. A similar response to weaker demand for AI-exposed skills could disproportionately affect new entrants and help explain the movement we observe into lower-paying sectors.

AI might also reduce firms' labor demand by affecting how workers acquire skills early in their careers. For example, entry-level tasks often serve both productive and training roles. If AI automates the work through which junior employees acquire tacit knowledge, firms may hire fewer new workers, reduce training, or reorganize career progression even when experienced workers benefit from the technology (Garicano and Rayo, 2025; Ide, 2025). These mechanisms could also disproportionately affect labor market entrants who have yet to accumulate skills and training on the job. The literature on recessions has shown that delayed employment or entry into lower-paying jobs can reduce the accumulation of relevant experience and place graduates on different career paths (Oreopoulos et al., 2012). If this is also true of the most AI-exposed graduates, then the negative outcomes we observe may be persistent.

One concern with explanations based on labor substitution is that impacts on early career workers seem to precede widespread AI adoption by firms (e.g., Bonney et al., 2024). One plausible explanation for this discrepancy is that AI may also affect the information available to employers when hiring new workers. Employers use observable markers of worker skill as signals of ability when choosing which workers to hire, and skill signals have the greatest impact on hiring when they are easily verified (Piopiunik et al., 2020). The widespread availability of generative AI to students may reduce the informativeness of observable signals like grades, assignments, and work samples if employers cannot distinguish underlying skill from AI-assisted performance. Consistent with this possibility, Hausman et al. (2026) and Chirikov (2026) both study variation in the AI exposure of university courses and find that AI availability raises grades, compressing the underlying grade distribution in ways that could reduce signal informativeness. Although higher grades could in principle reflect subject-specific learning gains, Chirikov (2026) also finds that grades increase the most in AI-exposed courses where homework is weighted more heavily, likely reflecting differences in students' ability to produce work with AI tools rather than differences in AI's ability to aid in learning.[32] The quick erosion of education signals leading to reduced hiring would be consistent with an effect of AI that preempts AI's widespread adoption by firms or use to alter firms' business processes (e.g., Bonney et al., 2024; Babina et al., 2024). It might also help explain the importance of graduation timing in our job switching results. Whether such a signaling issue might be persistent or transitory may depend on the ability of technological or pedagogical changes to more clearly distinguish skill acquisition from AI use.

In general, these labor demand channels do not imply that the negative changes must persist. Productivity gains may raise labor demand elsewhere, and our results show that workers adjust by moving into less-exposed tasks and sectors. Consistent with the possibility of worker adjustment, Hyman et al. (2025) find substantial returns to training among workers from AI-exposed occupations, driven primarily by transitions into less-exposed work. At the same time, productivity gains at adopting firms and the creation of new tasks may offset direct displacement by expanding output or creating demand for AI-complementary skills (Acemoglu and Restrepo, 2018; Autor et al., 2026). In the longer run, Kording and Marinescu (2025) emphasize that AI can generate offsetting productivity and substitution effects, raising output while reallocating labor away from automated cognitive tasks. At the same time, a technology-driven change in relative skill demand

[32]Studies at the secondary school level also find that positive effects of unrestricted AI availability on assignment performance correspond to negative effects in subsequent evaluations of learning (Bastani et al., 2025; Strömberg et al., 2026)

may be more persistent than a cyclical downturn because the original jobs or tasks may not return. Our short post period does not allow us to determine whether worker reallocation, retraining, productivity gains, and the creation of new tasks will ultimately offset the initial disruption we observe.

We outlined several ways in which generative AI could produce the patterns we observe, but other contemporaneous changes may also have contributed to the differential outcomes of exposed graduates. We consider several prominent alternatives, particularly addressing work-from-home, changes in graduate supply, and increases in self-employment or graduate school enrollment. These exercises do not readily explain the timing or concentration of our estimates, although our design cannot rule out other time-varying shocks to particular fields that coincide with the diffusion of generative AI. Taken together, the timing of the estimated changes, their concentration at the top of the exposure distribution, and the accompanying sectoral reallocation are consistent with an important role for generative AI in shaping the labor market entry of recent graduates.

The longer-run consequences remain uncertain. Increased job switching and attenuated earnings effects suggest that some exposed graduates adjust after initially entering weaker jobs. Whether these adjustments, together with retraining, productivity gains, and the creation of new tasks, are sufficient to offset the initial disruption or instead leave persistent effects on job matching and human capital accumulation remains to be seen.

9 Conclusion

This paper provides new evidence on how the rapid diffusion of generative AI has affected the labor market entry of recent college graduates. We assign AI exposure based on the pre-period occupational employment associated with a graduate's field of study and use linked employment histories to follow employment, earnings, employers, and industries after graduation. We find negative effects following the introduction of ChatGPT are concentrated among graduates from the most AI-exposed majors. These graduates are less likely to find employment immediately after graduation and experience substantial declines in earnings when employed. About half of the earnings decline reflects a shift away from high-paying industry sectors and toward lower-paying sectors such as Retail Trade and Accommodation and Food Services. The effects we estimate generally attenuate as workers move further from graduation. We also observe results related to job switching that suggest graduation timing may be a factor for exposed college majors.

These results suggest that the early labor market effects of generative AI are observable, but that new labor market entrants adjust to changes in labor demand by changing where they begin their careers. These adjustments may have outsized impacts for new graduates, whose first jobs provide both earnings and opportunities to demonstrate and accumulate skills. Our results do not identify a single mechanism behind these changes. Reduced demand for entry-level tasks, changes in training and hiring, and weaker educational signals may all contribute, and the long-run effects will depend on how firms, workers, and educational institutions adapt. The short post-ChatGPT period also limits what we can say about persistence. Even so, the evidence shows that highly exposed graduates have borne substantial costs in the initial adjustment to generative AI. Understanding whether these workers ultimately recover, and how the technology reshapes entry-level career paths and human capital accumulation, will be central to assessing the longer-run distributional consequences of AI.

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Appendix

A Additional Tables and Figures

Table A1: List of PSEO partners included in sample

Bryant University
The City University of New York
Connecticut State Colleges and Universities
Georgia Independent College Association
Illinois Board of Higher Education
Indiana Commission for Higher Education
Iowa Board of Regents
Louisiana Board of Regents
Massachusetts Department of Higher Education
Missouri Department of Higher Education and Workforce Development
Montana University System
New Jersey Office of the Secretary of Higher Education
New Mexico Higher Education Department
Ohio Department of Higher Education
Oklahoma State Regents for Higher Education
The Pennsylvania State University
South Carolina Commission on Higher Education
Tennessee Higher Education Commission
Texas Higher Education Coordinating Board
University of Hawai'i System
University of North Carolina System
University System of Georgia
University of Texas System
University of Wisconsin System
Utah System of Higher Education
West Virginia Higher Education Policy Commission
Western Governors University

Table A2: Characteristics of PSEO institutions

	Sample %	Overall %
Institution characteristics		
Public	92.7	53.4
Locale: City	67.0	61.0
Locale: Suburban	18.5	25.5
Size: Over 20,000	55.8	43.9
Carnegie: Doctoral Very High Research	43.2	32.6
Carnegie: Doctoral High Research	20.4	13.7
Carnegie: Master's Colleges (All Sizes)	26.7	27.1
Student characteristics		
Female	57.7	59.0
American Indian or Alaska Native	0.5	0.4
Asian	7.5	8.4
Black or African American	10.0	8.8
Hispanic	15.6	16.5
Native Hawaiian / Pacific Islander	0.1	0.2
White (Non-Hispanic)	57.0	52.4
Two or more races	3.7	4.0

Notes: Data from the Integrated Postsecondary Education Data System (IPEDS) 2022 "Institution Characteristics" and "Completions" surveys. The PSEO institution sample is identified by crosswalking OPEID to UnitID using information provided by College Scorecard. For privacy considerations and disclosure avoidance, student characteristics are derived using the counts of students in IPEDS alone, not the characteristics of students in the PSEO microdata. Not all levels of each variable are shown.

Table A3: AI Exposure by Field of Degree

Field of Degree	$N_{2018\text{-}22}$	Beta	Core	Usage	Share top	Employment		Earnings	
GPT-4 Beta Decile: 10									
2102: Computer Science	43,500	3.40	2.90	1.65	0.662	-0.061	(0.001)	-0.190	(0.003)
2407: Computer Engineering	18,500	3.17	2.56	1.18	0.598	-0.068	(0.002)	-0.120	(0.004)
2101: Computer Programming	6,400	2.60	2.11	1.91	0.543	-0.023	(0.003)	-0.113	(0.008)
2105: Information Sciences	18,000	2.15	2.27	1.61	0.523	-0.011	(0.002)	-0.079	(0.007)
2100: Computer and Information Systems-General	63,000	2.07	2.19	1.39	0.556	-0.054	(0.002)	-0.142	(0.003)
6212: Management Information Systems and Statistics	27,500	1.90	2.27	1.51	0.483	-0.021	(0.001)	-0.089	(0.003)
2106: Computer Administration Management and Security	14,000	1.88	2.34	1.57	0.545	-0.033	(0.003)	-0.096	(0.008)
3701: Applied Mathematics	2,700	1.74	1.57	1.33	0.383	-0.011	(0.003)	-0.101	(0.008)
3702: Statistics	3,900	1.61	1.60	1.41	0.393	-0.026	(0.002)	-0.081	(0.006)
2408: Electrical Engineering	27,000	1.48	1.10	-0.39	0.267	-0.011	(0.001)	0.016	(0.003)
6202: Actuarial Science	1,900	1.41	1.19	0.04	0.259	-0.014	(0.003)	0.113	(0.007)
3700: Mathematics	22,500	1.09	0.98	0.71	0.327	-0.015	(0.001)	-0.051	(0.003)
2401: Aerospace Engineering	7,400	1.08	0.59	-1.02	0.174	-0.032	(0.003)	-0.091	(0.007)
6205: Business Economics	6,100	1.05	1.17	1.36	0.284	-0.038	(0.002)	0.020	(0.006)
2107: Computer Networking and Telecommunications	2,300	1.04	1.05	0.89	0.426	-0.071	(0.004)	-0.056	(0.010)
1902: Journalism	21,500	1.02	0.97	0.99	0.357	-0.011	(0.002)	-0.066	(0.005)
GPT-4 Beta Decile: 9									
6201: Accounting	82,500	0.98	1.26	1.77	0.162	-0.001	(0.001)	0.031	(0.002)
6207: Finance	75,500	0.98	1.21	1.87	0.276	-0.021	(0.001)	-0.009	(0.004)
5501: Economics	31,000	0.93	1.18	1.33	0.285	-0.001	(0.001)	-0.052	(0.004)
5599: Miscellaneous Social Sciences	1,000	0.93	0.71	0.37	0.268	0.067	(0.004)	-0.029	(0.015)
2409: Engineering Mechanics, Physics, and Science	1,700	0.92	1.06	-0.44	0.230	-0.012	(0.004)	-0.008	(0.009)
5007: Physics	7,700	0.85	0.66	0.03	0.249	-0.027	(0.001)	-0.104	(0.004)
6204: Operations, Logistics and E-Commerce	18,000	0.80	1.09	0.37	0.284	-0.019	(0.002)	-0.002	(0.005)
2412: Industrial and Manufacturing Engineering	13,500	0.80	1.15	-0.40	0.195	-0.019	(0.002)	0.021	(0.005)
6099: Miscellaneous Fine Arts	2,700	0.77	0.31	0.61	0.324	-0.041	(0.004)	-0.105	(0.009)
1904: Advertising and Public Relations	23,000	0.72	0.89	1.48	0.262	-0.032	(0.002)	-0.062	(0.005)
2404: Biomedical Engineering	12,500	0.71	0.76	-0.39	0.212	-0.030	(0.002)	-0.083	(0.005)
2418: Nuclear Engineering	1,400	0.65	0.47	-1.10	0.131	-0.052	(0.003)	0.102	(0.009)
2405: Chemical Engineering	18,500	0.61	0.91	-1.14	0.120	0.000	(0.002)	0.039	(0.005)
GPT-4 Beta Decile: 8									
6206: Marketing	83,000	0.59	0.58	1.51	0.258	-0.003	(0.001)	-0.045	(0.003)
6210: International Business	4,700	0.57	0.63	1.04	0.263	-0.004	(0.002)	-0.034	(0.008)
2402: Biological Engineering	3,600	0.57	0.71	-0.79	0.173	0.038	(0.003)	0.103	(0.007)
1903: Mass Media	31,000	0.54	0.17	0.62	0.274	-0.018	(0.002)	-0.069	(0.003)
2502: Electrical Engineering Technology	2,700	0.54	0.45	-0.25	0.236	-0.006	(0.003)	0.117	(0.006)
5504: Geography	6,200	0.53	0.47	-0.02	0.280	0.009	(0.001)	0.117	(0.003)
5402: Public Policy	1,300	0.53	0.81	0.68	0.234	-0.060	(0.005)	-0.045	(0.010)
2414: Mechanical Engineering	59,000	0.53	0.59	-1.21	0.110	-0.009	(0.001)	0.039	(0.003)

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Table A3 – continued from previous page

Field of Degree	$N_{2018\text{-}22}$	Beta	Core	Usage	Share top	Employment		Earnings	
5505: International Relations	4,600	0.53	0.83	0.64	0.268	0.020	(0.003)	0.008	(0.006)
2419: Petroleum Engineering	3,600	0.51	0.25	-1.18	0.139	0.038	(0.004)	0.047	(0.007)
5002: Atmospheric Sciences and Meteorology	1,400	0.48	0.54	-1.01	0.146	-0.004	(0.003)	-0.143	(0.006)
6103: Health and Medical Administrative Services	23,000	0.47	0.41	0.65	0.277	0.009	(0.003)	0.038	(0.005)
6299: Miscellaneous Business	14,000	0.46	0.51	0.91	0.237	0.008	(0.002)	-0.015	(0.004)
GPT-4 Beta Decile: 7									
6203: Business Management and Administration	158,000	0.46	0.44	0.79	0.257	-0.009	(0.001)	-0.005	(0.003)
2413: Materials Engineering	4,500	0.42	0.80	-0.98	0.139	-0.014	(0.003)	0.026	(0.008)
2400: General Engineering	2,600	0.40	0.40	-0.77	0.175	0.050	(0.006)	0.051	(0.017)
1901: Communications	64,000	0.39	0.41	0.87	0.257	-0.003	(0.001)	-0.044	(0.002)
6209: Human Resources and Personnel Management	17,500	0.37	0.44	0.19	0.192	0.007	(0.002)	-0.015	(0.005)
5506: Political Science and Government	44,000	0.36	0.59	0.57	0.253	-0.003	(0.001)	-0.011	(0.002)
2410: Environmental Engineering	2,400	0.35	0.58	-1.56	0.079	0.028	(0.002)	0.022	(0.006)
3302: Composition and Speech	6,000	0.33	0.21	0.45	0.270	0.016	(0.003)	-0.001	(0.006)
GPT-4 Beta Decile: 6									
2500: Engineering Technologies	5,500	0.29	0.40	-0.16	0.258	0.024	(0.003)	0.041	(0.006)
6200: General Business	45,000	0.29	0.34	0.68	0.255	-0.009	(0.003)	0.014	(0.007)
2603: Other Foreign Languages	2,100	0.28	0.19	0.40	0.247	-0.016	(0.003)	0.054	(0.007)
5401: Public Administration	6,300	0.27	0.49	0.13	0.252	-0.010	(0.004)	-0.035	(0.008)
2499: Energy Systems Engineering	3,700	0.26	0.18	-1.15	0.115	0.014	(0.002)	0.063	(0.006)
3202: Pre-Law and Legal Studies	3,500	0.22	0.42	0.70	0.211	0.004	(0.003)	0.044	(0.010)
3301: English Language and Literature	45,500	0.22	0.17	0.47	0.257	-0.005	(0.001)	-0.028	(0.002)
2599: Energy Systems Technologies/Technicians	15,000	0.20	0.14	-0.87	0.207	-0.008	(0.002)	0.067	(0.004)
1401: Architecture	14,000	0.19	-0.15	-1.20	0.100	-0.006	(0.002)	-0.015	(0.005)
2403: Architectural Engineering	1,200	0.18	0.13	-1.39	0.119	-0.006	(0.004)	0.089	(0.009)
2406: Civil Engineering	25,500	0.16	-0.05	-1.81	0.059	0.033	(0.001)	0.077	(0.003)
6006: Art History and Criticism	1,800	0.15	0.19	0.49	0.235	0.047	(0.002)	-0.007	(0.006)
6004: Commercial Art and Graphic Design	21,500	0.14	-0.67	0.44	0.136	-0.004	(0.002)	-0.070	(0.005)
1501: Area, Ethnic, Cultural, Gender, and Group Studies	5,900	0.14	0.18	0.35	0.234	-0.002	(0.002)	-0.028	(0.006)
2601: Linguistics and Comparative Language and Literature	5,500	0.10	-0.04	0.22	0.226	0.006	(0.002)	0.018	(0.004)
4001: Intercultural and International Studies	5,700	0.10	0.17	0.33	0.215	-0.034	(0.003)	-0.030	(0.006)
2602: French, German, Latin and Other Language Studies	9,300	0.08	0.10	0.21	0.218	0.025	(0.001)	0.011	(0.004)
2001: Communication Technologies	2,700	0.01	-0.57	0.28	0.214	0.032	(0.004)	-0.032	(0.009)
1102: Agricultural Economics	3,800	-0.01	-0.14	-0.11	0.201	-0.048	(0.004)	0.013	(0.009)
1199: Miscellaneous Agriculture	1,900	-0.02	0.07	0.44	0.225	0.037	(0.003)	-0.021	(0.006)
3611: Neuroscience	7,500	-0.04	-0.15	-0.44	0.160	0.009	(0.003)	0.031	(0.006)
6005: Film, Video and Photographic Arts	9,000	-0.05	-0.75	-0.03	0.203	0.024	(0.004)	-0.046	(0.007)
5502: Anthropology and Archeology	9,800	-0.08	-0.06	0.07	0.200	-0.017	(0.001)	-0.034	(0.003)
1104: Food Science	2,300	-0.08	0.16	-0.63	0.128	-0.040	(0.003)	0.047	(0.008)
5004: Geology and Earth Science	8,300	-0.09	-0.10	-0.77	0.133	-0.040	(0.001)	0.010	(0.004)
GPT-4 Beta Decile: 5									

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Table A3 – continued from previous page

Field of Degree	$N_{2018\text{-}22}$	Beta	Core	Usage	Share top	Employment		Earnings	
3603: Molecular Biology	4,900	-0.10	-0.14	-0.60	0.137	0.003	(0.004)	-0.065	(0.009)
2504: Mechanical Engineering Related Technologies	6,100	-0.11	0.11	-1.51	0.122	-0.010	(0.003)	0.048	(0.007)
4801: Philosophy and Religious Studies	7,200	-0.12	-0.08	-0.04	0.194	-0.008	(0.001)	0.130	(0.004)
3601: Biochemical Sciences	13,000	-0.17	-0.20	-0.48	0.136	0.018	(0.001)	-0.044	(0.004)
2503: Industrial Production Technologies	7,100	-0.17	0.09	-1.36	0.136	0.018	(0.003)	0.026	(0.007)
5003: Chemistry	20,500	-0.20	-0.12	-0.49	0.122	-0.014	(0.001)	0.030	(0.003)
5507: Sociology	34,000	-0.22	-0.17	-0.06	0.200	0.014	(0.001)	0.033	(0.002)
6402: History	31,500	-0.24	-0.18	-0.03	0.202	0.004	(0.001)	0.001	(0.002)
6110: Community and Public Health	30,500	-0.25	-0.22	-0.10	0.189	0.009	(0.002)	0.004	(0.004)
6211: Hospitality Management	14,000	-0.28	-0.31	-0.29	0.153	0.000	(0.002)	0.012	(0.004)
3606: Microbiology and Immunology	4,600	-0.28	-0.30	-0.52	0.155	0.005	(0.002)	0.056	(0.005)
1301: Environmental Science	14,500	-0.30	-0.09	-0.58	0.156	-0.004	(0.001)	0.005	(0.003)
6000: Fine Arts	16,500	-0.33	-0.76	0.04	0.168	0.006	(0.001)	-0.007	(0.003)
1101: Agriculture Production and Management	19,000	-0.33	-0.44	-0.44	0.217	0.009	(0.002)	0.047	(0.005)
6002: Music	20,000	-0.34	-0.68	-0.47	0.166	0.013	(0.001)	-0.014	(0.003)
2501: Engineering and Industrial Management	2,600	-0.36	-0.02	-0.80	0.108	-0.012	(0.005)	0.064	(0.009)
4007: Interdisciplinary Social Sciences	4,500	-0.36	-0.32	-0.26	0.149	-0.007	(0.002)	0.034	(0.006)
3605: Genetics	1,900	-0.37	-0.51	-0.83	0.094	-0.004	(0.004)	0.020	(0.007)
3699: Miscellaneous Biology	6,100	-0.41	-0.44	-0.74	0.134	-0.024	(0.003)	0.017	(0.006)
5500: General Social Sciences	4,600	-0.41	-0.45	-0.22	0.205	0.030	(0.004)	-0.043	(0.013)
5299: Miscellaneous Psychology	4,100	-0.43	-0.38	-0.34	0.155	0.012	(0.008)	-0.001	(0.013)
GPT-4 Beta Decile: 4									
6108: Pharmacy, Pharmaceutical Sciences, and Admin.	3,400	-0.44	-0.70	-1.01	0.087	0.050	(0.006)	0.064	(0.009)
3600: Biology	129,000	-0.48	-0.51	-0.42	0.164	0.004	(0.001)	0.001	(0.002)
3402: Humanities	1,900	-0.49	-0.44	-0.30	0.192	0.071	(0.005)	0.032	(0.011)
5200: Psychology	165,000	-0.51	-0.43	-0.18	0.180	0.008	(0.001)	0.018	(0.002)
GPT-4 Beta Decile: 3									
5098: Multi-disciplinary or General Science	70,000	-0.52	-0.59	-0.41	0.170	0.003	(0.002)	-0.009	(0.005)
6007: Studio Arts	15,000	-0.54	-0.84	-0.08	0.159	0.013	(0.002)	-0.069	(0.004)
5503: Criminology	12,000	-0.55	-0.47	-0.42	0.190	-0.003	(0.002)	0.070	(0.007)
4002: Nutrition Sciences	5,300	-0.55	-0.89	-0.62	0.111	0.009	(0.003)	0.048	(0.005)
5403: Human Services and Community Organization	3,800	-0.60	-0.63	-0.44	0.151	-0.038	(0.004)	0.021	(0.013)
3401: Liberal Arts	78,500	-0.67	-0.58	-0.36	0.163	-0.005	(0.001)	-0.010	(0.004)
3608: Physiology	8,500	-0.74	-0.75	-0.47	0.143	0.025	(0.003)	0.018	(0.007)
6106: Health and Medical Preparatory Programs	2,600	-0.74	-0.79	-0.42	0.144	0.042	(0.007)	-0.120	(0.015)
1303: Natural Resources Management	8,800	-0.75	-0.57	-1.03	0.120	0.005	(0.002)	0.063	(0.006)
6001: Drama and Theater Arts	14,000	-0.75	-0.80	-0.32	0.164	-0.010	(0.002)	0.019	(0.004)
6003: Visual and Performing Arts	6,200	-0.77	-0.87	-0.40	0.141	0.017	(0.003)	0.062	(0.006)
2901: Family and Consumer Sciences	41,000	-0.79	-0.74	-0.54	0.139	0.018	(0.001)	0.041	(0.004)
5404: Social Work	38,000	-0.80	-0.72	-0.54	0.142	0.025	(0.002)	0.047	(0.003)
2399: Miscellaneous Education	12,500	-0.80	-0.69	-0.42	0.097	0.034	(0.004)	0.061	(0.009)

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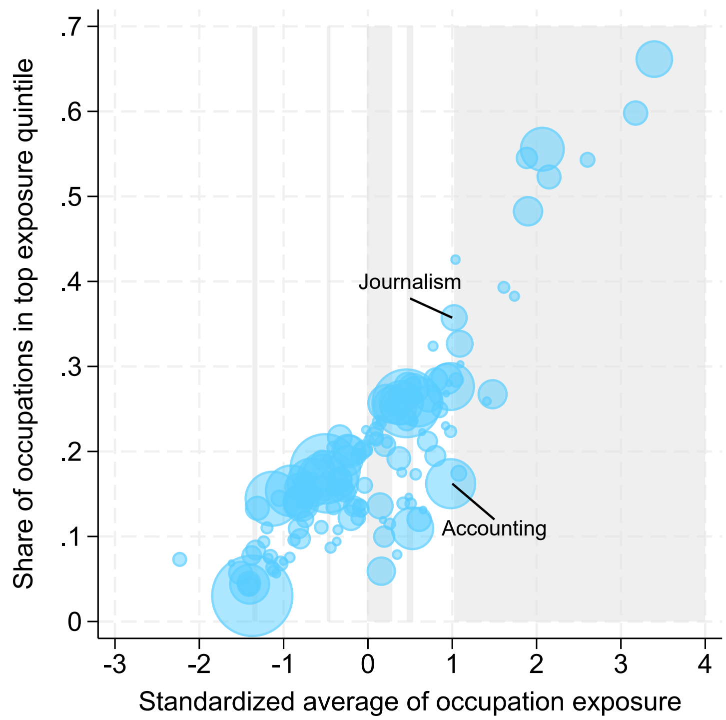
Table A3 – continued from previous page

Field of Degree	$N_{2018\text{-}22}$	Beta	Core	Usage	Share top	Employment		Earnings	
6199: Miscellaneous Health Medical Professions	12,500	-0.82	-1.22	-0.89	0.109	0.011	(0.002)	-0.066	(0.005)
6102: Communication Disorders Sciences and Services	21,500	-0.83	-0.83	-0.62	0.136	0.018	(0.001)	0.048	(0.003)
GPT-4 Beta Decile: 2									
6100: General Medical and Health Services	21,500	-0.83	-0.76	-0.60	0.159	0.018	(0.002)	0.014	(0.004)
3604: Ecology	4,200	-0.88	-0.78	-0.99	0.096	-0.025	(0.004)	0.066	(0.011)
5301: Criminal Justice and Fire Protection	88,000	-0.91	-0.80	-0.76	0.154	0.011	(0.002)	0.050	(0.004)
2311: Social Science or History Teacher Education	3,000	-0.93	-0.84	-0.58	0.075	0.019	(0.002)	-0.035	(0.006)
2308: Science Teacher Education	1,500	-1.00	-0.93	-0.80	0.071	-0.023	(0.004)	0.030	(0.008)
2314: Art and Music Education	6,400	-1.04	-1.05	-0.68	0.068	0.019	(0.002)	0.005	(0.005)
5901: Transportation Sciences and Technologies	7,900	-1.05	-1.38	-1.26	0.145	0.025	(0.003)	-0.044	(0.009)
2305: Mathematics Teacher Education	2,000	-1.09	-0.95	-0.31	0.056	0.039	(0.003)	0.066	(0.007)
2313: Language and Drama Education	3,500	-1.12	-0.99	-0.54	0.059	-0.002	(0.002)	-0.080	(0.008)
4101: Physical Fitness, Parks, Recreation, and Leisure	100,000	-1.13	-0.96	-0.73	0.144	0.005	(0.001)	-0.016	(0.002)
2309: Secondary Teacher Education	5,000	-1.13	-0.95	-0.54	0.063	0.047	(0.002)	0.049	(0.007)
2300: General Education	6,200	-1.16	-1.07	-0.69	0.076	0.024	(0.003)	-0.012	(0.007)
1302: Forestry	2,100	-1.18	-1.30	-1.71	0.075	0.009	(0.004)	0.015	(0.011)
3609: Zoology	4,000	-1.20	-1.06	-1.22	0.110	0.017	(0.003)	0.124	(0.010)
1105: Plant Science and Agronomy	4,100	-1.23	-1.08	-1.28	0.094	-0.003	(0.002)	-0.025	(0.005)
1103: Animal Sciences	18,000	-1.31	-1.09	-1.08	0.133	0.017	(0.002)	0.068	(0.005)
2306: Physical and Health Education Teaching	12,000	-1.33	-1.15	-0.78	0.084	0.020	(0.003)	0.030	(0.006)
GPT-4 Beta Decile: 1									
6107: Nursing	222,000	-1.37	-1.50	-1.74	0.030	0.028	(0.002)	0.039	(0.005)
6109: Treatment Therapy Professions	11,500	-1.38	-1.29	-1.33	0.078	-0.001	(0.003)	0.014	(0.007)
2304: Elementary Education	52,000	-1.40	-1.31	-0.85	0.044	0.019	(0.001)	0.020	(0.002)
2312: Teacher Education: Multiple Levels	10,500	-1.41	-1.34	-0.82	0.041	0.008	(0.002)	0.065	(0.006)
6105: Medical Technologies Technicians	17,000	-1.41	-1.41	-1.82	0.045	0.030	(0.002)	0.068	(0.006)
2310: Special Needs Education	16,000	-1.41	-1.13	-1.25	0.043	0.020	(0.002)	0.081	(0.004)
2307: Early Childhood Education	17,500	-1.51	-1.38	-1.13	0.057	0.015	(0.002)	0.051	(0.005)
2301: Educational Administration and Supervision	1,000	-1.62	-1.26	-1.68	0.069	-0.001	(0.006)	0.110	(0.016)
6104: Medical Assisting Services	5,600	-2.23	-2.13	-2.06	0.073	0.037	(0.003)	0.051	(0.008)

Note: "N" denotes the number of graduates between 2018-2022 who majored in the specified field. "Beta" is the Eloundou et al., GPT-4 Beta score. "Core" is the Eisfeldt et al. core task exposure score. "Usage" is the Handa et al. observed task-usage score. All three scores are individually standardized to be mean zero with a variance of one. "Share top" is the share of a major's occupations in the top quintile of occupation-level exposure defined by the GPT-4 Beta score. "Employment" and "Earnings" are the major-specific coefficients (centered at zero) and standard errors (clustered by major) for the year 2024 (with 2022 as the reference year) from the event study in Equation (2) with the employment rate and log full-quarter earnings from dominant job as outcomes. Horizontal lines mark the decile cutoffs based on the GPT-4 Beta score. Only majors with at least 1,000 graduates are included.

Figure A1 plots each major by its exposure using the weighted mean of occupation exposure on the x axis, and the share of each major's occupations that are in the top quintile of exposure on the y axis. The two measures are highly correlated, but there is substantial variation in mean exposure and top-quintile share, especially in some middle deciles. For instance, Journalism and Accounting have similar mean exposure measures, but Journalism has a much higher share of occupations in the highest quintile of the Eloundou et al. exposure measure.

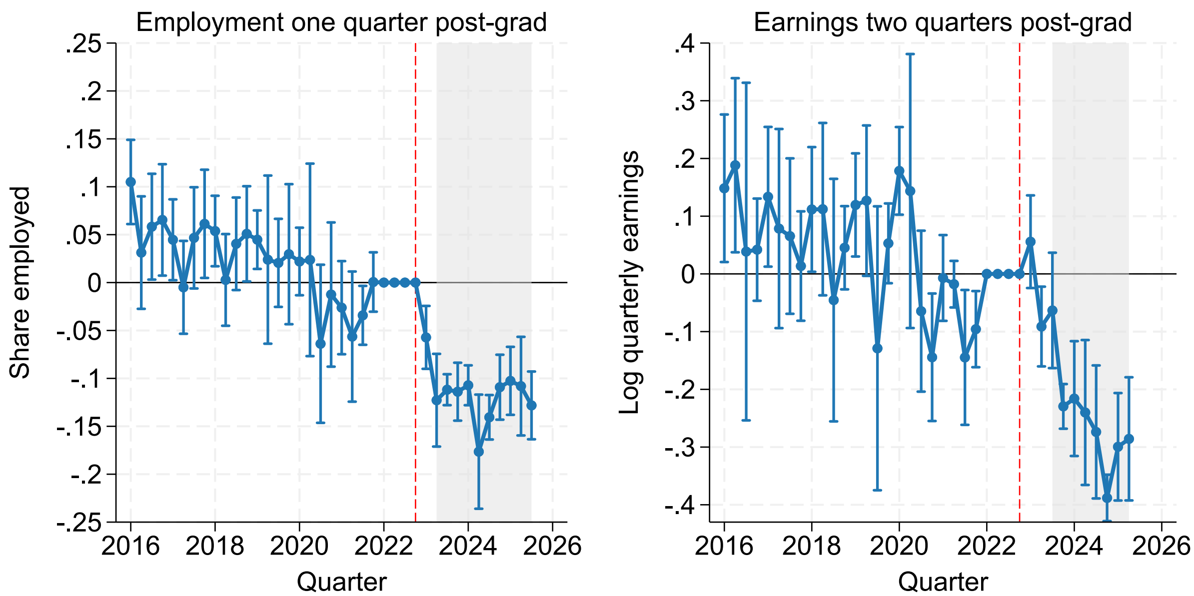
Figure A1: Comparing Eloundou exposure specifications



Notes: Weighted average of occupation exposure is the Eloundou GPT-4 occupation-level score averaged over each college major's occupation distribution from the ACS. Share of occupations in the top exposure quintile is the percent of a major's employed graduates with an occupation-level score above 0.567. Bubble sizes correspond to count of graduates from 2018 to 2022. Only majors with at least 1,000 graduates are plotted. Shaded regions denote alternating deciles of exposure.

Figure A2 re-estimates our baseline event-study specification using the share of each major's occupations in the top quintile of the GPT-4 Beta occupation-level exposure score as the explanatory variable. This specification uses variation across all majors to examine whether labor market outcomes deteriorate more for fields whose graduates are concentrated in the most AI-exposed occupations. Callaway et al. (Forthcoming) show that a two-way fixed effects model with continuous treatment, such as the one estimated here, cannot recover the true global average causal response, but instead recovers a weakly causal weighted average of treatment effects under a stronger parallel trends assumption. Any interpretation of these coefficients should be done with caution.

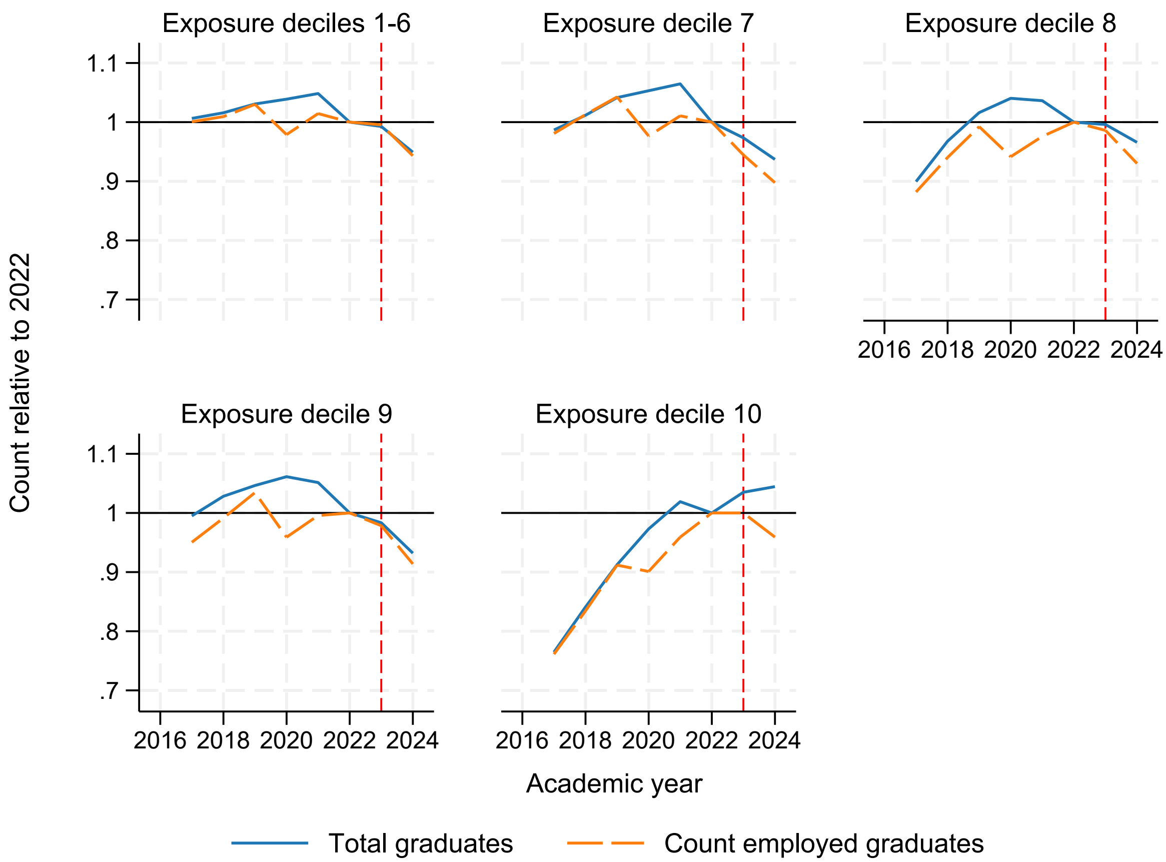
Figure A2: Regressions on share of occupations in top quintile



Notes: Coefficients correspond to effects of moving from no employment share in the top exposure quintile of occupations to 100% employment share in the top quintile of occupations based on Eloundou et al. (2024). Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. Estimates from a regression with institution-major, institution-quarter, and major-quarter of year fixed effects. Quarter denotes observation period. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 5,685,000 in employment regression and 3,225,000 in earnings regression. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

Figure A3 plots the count of graduates by GPT-4 Beta exposure decile and the count of those graduates employed one quarter after graduation. Both counts are relative to 2022. While we use the count of graduates in PSEO to measure growth in each major, the figure is very similar when using national graduate counts from the IPEDS surveys. We aggregate to academic year and index to 2022 to permit comparisons with IPEDS.

Figure A3: Supply and employment of graduates relative to 2022

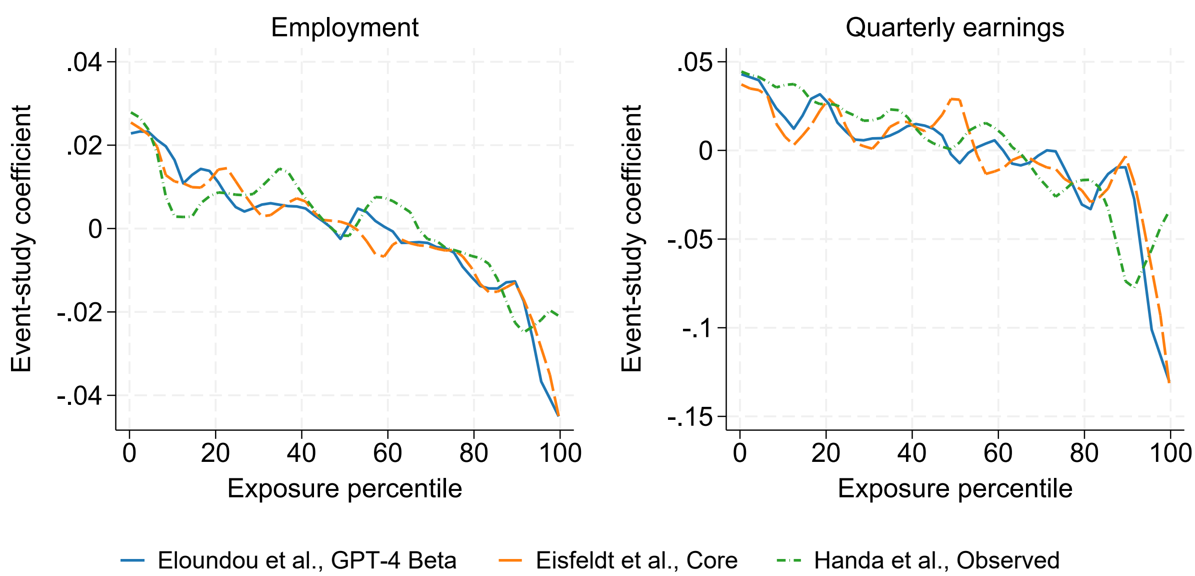


Notes: Figure contains count of graduates in PSEO by Eloundou et al. GPT-4 Beta exposure decile relative to 2022 value (dark blue) and count of employed graduates one quarter post-graduation relative to 2022 value (light blue). November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. Number of observations: 4,821,600. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

B Alternative sources of AI exposure

In Section 2.2, we contrasted our chosen AI exposure measure, Eloundou et al.'s potential task performance ratings as scored by GPT-4, with those derived from actual usage statistics, such as the measure of Handa et al. (2025). We also alluded to another measure, developed by Eisfeldt et al. (2023), that is also exposure-based but has the highest correlation with current industry-level adoption of AI (Tucker, 2026). We use our major-level event study regressions to evaluate how closely the outcome relationships estimated using these three measures align. Figure A4 presents the local polynomial-smoothed trend between each major's exposure percentile, separately by source, and major-specific event-study coefficients for the change in initial employment or initial earnings from 2022 to 2024.

Figure A4: Major-level event study coefficients by different AI measures

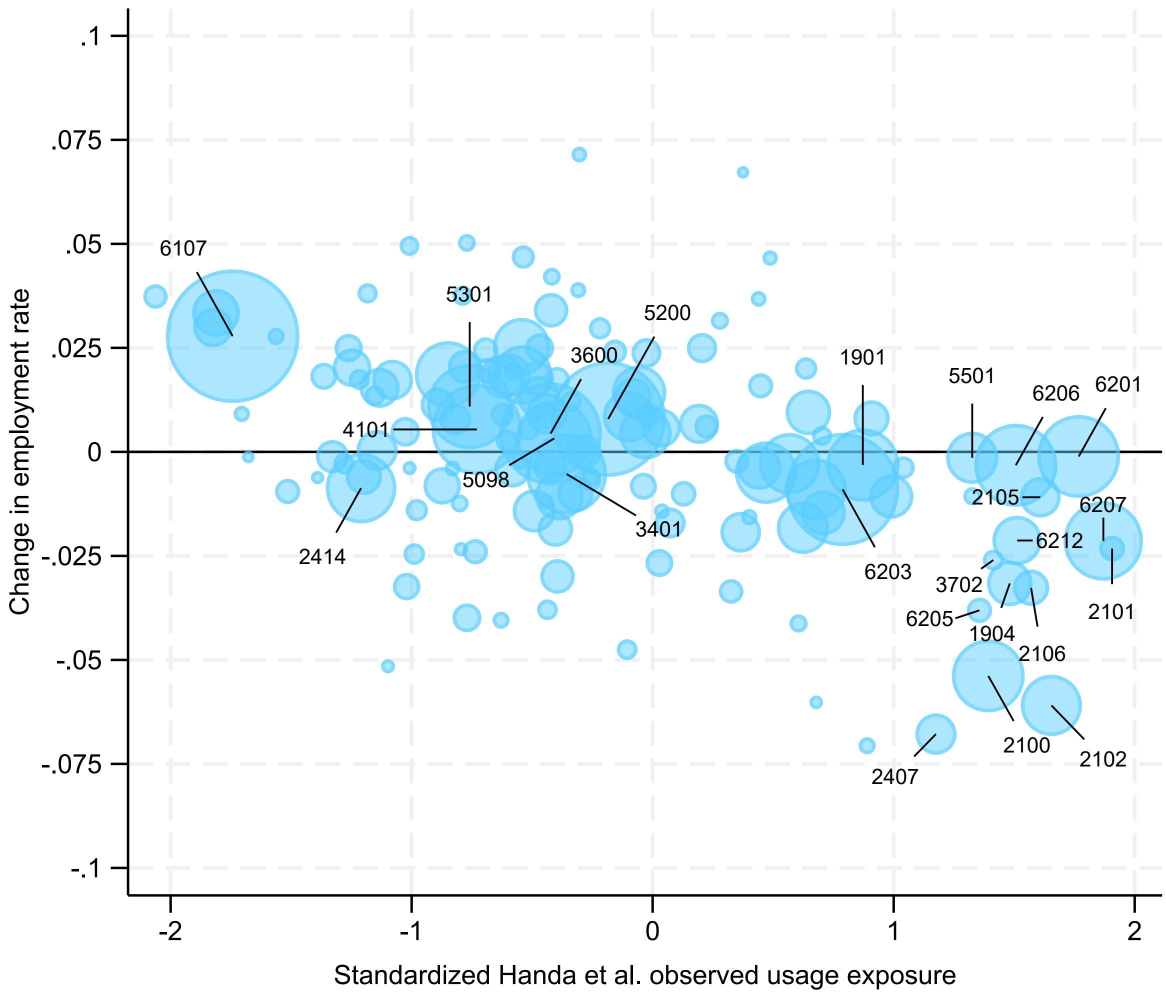


Notes: Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. Estimates based on a regression of employment or earnings on major-year coefficients with institution-major, institution-quarter, and major-quarter of year fixed effects. Reference year 2022. Coefficients re-centered relative to overall year average. Only coefficients for the year 2024 and majors with at least 1,000 graduates between 2018 and 2022 are included. A local polynomial fit with a Epanechnikov kernel and bandwidth of three plotted for each exposure percentile. Number of observations: 5,685,000 in employment regression and 3,225,000 in earnings regression. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

Through the eightieth percentile, the reordering of majors caused by varying exposure measurements does not change the observed relationship between exposure percentile and initial labor market declines. Around the ninetieth percentile of exposure defined by Eloundou et al. or Eisfeldt et al., the estimated event-study coefficients are substantially lower than the trend that would be extrapolated from the less exposed majors. By contrast, the smoothed polynomial based on the Handa exposure measure decreases around the ninetieth percentile but rebounds to the trend established by less exposed majors. The cause for this can be seen in Appendix Figures A5 and A6 where the major-specific event-study coefficients are plotted by the standardized Handa exposure score. The Eloundou et al. and Eisfeldt et al. measures both order business majors like Accounting and Finance as less exposed than computer science majors, and the largest observed labor market declines are in these computer science fields. Handa swaps this ordering.

One nice feature of the Handa measure is that the authors incorporate additional task-level details intended to distinguish between automative or augmentative usage of AI. If the augmentative/automative split of the tasks performed via AI on behalf of majors differs, then we might expect different effects corresponding to the labor complementarity or substitution corresponding to that usage. The majors that are primarily exposed to AI's augmentative usage might benefit from the exposure, owing to increased productivity and the necessity of their continued human input, while the majors that are primary exposed to automative usage might be more vulnerable to displacement or reduced earnings. While this story of underlying heterogeneity is compelling, our analysis finds little evidence to support it. Figure A7 plots the share of AI prompts classified as "automative" in Handa et al.'s usage data against their standardized exposure score. There is a wide dispersion in the share of automative prompts at the low end of the exposure distribution, but this is partially a result of fewer prompts available to be classified. The top half of the exposure distribution has a tight band of automative shares between 0.33 and 0.41, and narrower ranges of exposure have an even tighter range of automative shares. For descriptive purposes, business majors (high exposure, small labor market effects) and computer & information systems majors (high exposure, large labor market effects) are highlighted. There is no observable difference in the share of prompts that are automative for business majors versus computer & information systems majors, and across all majors, there is no significant relationship between the share of prompts that are automative and labor market outcomes after controlling for exposure.

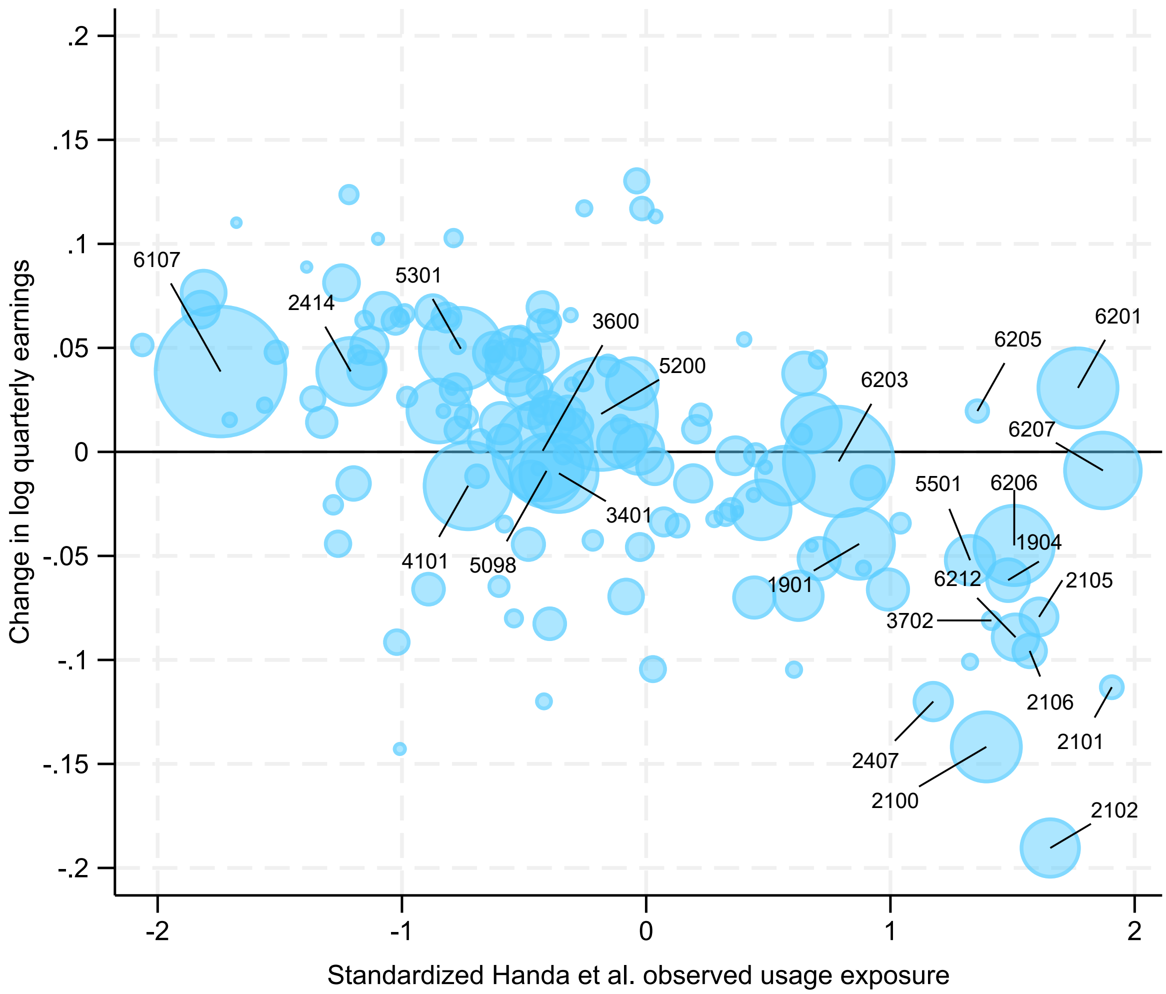
Figure A5: Share employed one-quarter post-graduation



1901: Communications
1904: Advertising and Public Relations
2100: Computer and Info. Systems-General
2101: Computer Programming
2102: Computer Science
2105: Info. Sciences
2106: Computer Admin. Mgmt. and Security
2407: Computer Engineering
2414: Mechanical Engineering
3401: Liberal Arts
3600: Biology
3702: Statistics
4101: Physical Fitness, Parks, Rec., and Leisure
5098: Multi-disciplinary or General Science
5200: Psychology
5301: Criminal Justice and Fire Protection
5501: Economics
6107: Nursing
6201: Accounting
6203: Business Mgmt. and Admin.
6205: Business Economics
6206: Marketing
6207: Finance
6212: Mgmt. Info. Systems and Statistics

Notes: Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Estimates based on a regression of employment on major-year coefficients with institution-major, institution-quarter, and major-quarter of year fixed effects. Reference year 2022. Coefficients re-centered relative to overall year average. Bubble sizes correspond to count of graduates from 2018 to 2022. Only coefficients for the year 2024 and majors with at least 1,000 graduates between 2018 and 2022 are plotted. Shaded regions denote alternating deciles of exposure. Number of observations: 5,685,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

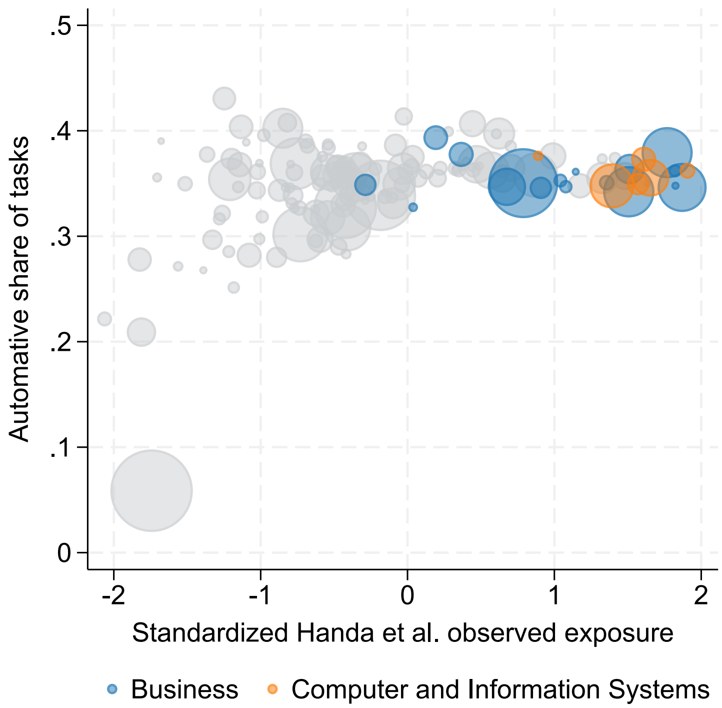
Figure A6: Log quarterly earnings two-quarters post-graduation



1901: Communications	2414: Mechanical Engineering	5501: Economics
1904: Advertising and Public Relations	3401: Liberal Arts	6107: Nursing
2100: Computer and Info. Systems-General	3600: Biology	6201: Accounting
2101: Computer Programming	3702: Statistics	6203: Business Mgmt. and Admin.
2102: Computer Science	4101: Physical Fitness, Parks, Rec., and Leisure	6205: Business Economics
2105: Info. Sciences	5098: Multi-disciplinary or General Science	6206: Marketing
2106: Computer Admin. Mgmt. and Security	5200: Psychology	6207: Finance
2407: Computer Engineering	5301: Criminal Justice and Fire Protection	6212: Mgmt. Info. Systems and Statistics

Notes: Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. Estimates based on a regression of earnings on major-year coefficients with institution-major, institution-quarter, and major-quarter of year fixed effects. Reference year 2022. Coefficients re-centered relative to overall year average. Bubble sizes correspond to count of graduates from 2018 to 2022. Only coefficients for the year 2024 and majors with at least 1,000 graduates between 2018 and 2022 are plotted. Shaded regions denote alternating deciles of exposure. Number of observations: 3,225,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

Figure A7: Share of automative tasks by major



Notes: Standardized Handa exposure is Handa et al. occupation-level score averaged over each college major's occupation distribution from the ACS. Automative share of tasks is the percent of a major's prompts, calculated by averaging over the major's occupation-level shares, classified as "directive" or "feedback loop". Bubble sizes correspond to count of graduates from 2018 to 2022. Only majors with at least 1,000 graduates are plotted.

C Work-from-home

We construct work-from-home rates as the fraction of employed bachelor degree holder's aged 21-29 that answered the question "how did [you] usually get to work last week" with "worked from home" in the ACS. Responses are aggregated by field of degree and survey year and weighted by ACS person weights. This measure is highly correlated with the Eloundou et al. GPT-4 Beta AI exposure score, as demonstrated in Figure A8. The large positive correlation between exposure and work-from-home is visible throughout the entire exposure distribution.

Figure A8: Correlation of work-from-home and AI exposure

Notes: Eloundou et al. GPT-4 Beta occupation-level score averaged over each college major's occupation distribution from the ACS to create major-level score. Work-from-home rates are the fraction of employed bachelor's degree holders aged 21-29 in the ACS that answered the question "how did [you] usually get to work last week" with "worked from home", aggregated by major and weighted by ACS person weights. Bubble sizes correspond to count of graduates from 2018 to 2022. Only majors with at least 1,000 graduates are plotted. Shaded regions denote alternating deciles of exposure. Number of observations in ACS: 83,064.

In our first specification, we allow the work-from-home rate to vary by year and estimate the effect of work-from-home using both variation across occupations and variation over time.[33,34]

[33]The 2025 ACS PUMS were not available at the time of writing this manuscript, so we used 2024 work-from-home rates for 2025.

[34]We considered allowing for geographic variation in work-from-home to allow for time-varying coefficients on its effects, but the sample sizes in the ACS are too small to consistently measure work-from-home rates by major, year,

The precise estimating equation is,

$$y_{it} = \sum_{s \notin 2022} \sum_{d > 6} \beta_{ds} 1[E(m(i)) = d, t = s] + \gamma W(m(i), t) + \delta_{c(i),m(i)} + \phi_{c(i),t} + \tau_{m(i),q(t)} + \varepsilon_{it}. \quad (3)$$

where $W(m(i),t)$ is the work-from-home rate for major $m(i)$ in time period t. Figures A9 and A10 present the results of this specification for outcomes of initial employment and initial earnings, respectively, along with the previously estimated baseline results for comparison.

Figure A9 shows that even when work-from-home is included in the regression, overall patterns of employment look fairly similar. The coefficients in the pre-period uniformly decrease when the additional covariate is added, but, with the exception of decile nine, they remain consistently indistinct from zero. In the post-period, none of the coefficients are practically or statistically different from their values in the baseline model without work-from-home controls. The coefficient on work-from-home is -0.013 with a standard error of 0.0022. Similarly, Figure A10 shows that estimated event study coefficients of AI exposure on initial earnings are indistinguishable from their baseline values in both the pre- and post-periods. Here, the coefficient on work-from-home is -0.0065 with a standard error of 0.0047. Overall, these results provide little evidence to suggest that our estimates are being driven by the rise of remote work. Even when compared to pre-pandemic averages, initial employment rates and initial earnings still appear to have declined after 2022, especially among the most-exposed decile.

One concern with the specification of Equation 3, time-constant AI exposure modeled with time-varying effects and time-varying work-from-home restricted to a time-invariant effect, is any changes over time in the relationship between work-from-home and employment/earnings will incorrectly be attributed to AI exposure. This motivates our second robustness check, which is a replication of Lambert and Schindler (2026). Major-level work-from-home rates are fixed at their 2022 values, and both AI exposure and work-from-home are standardized and treated as continuous variables with separate event study series. Specifically,

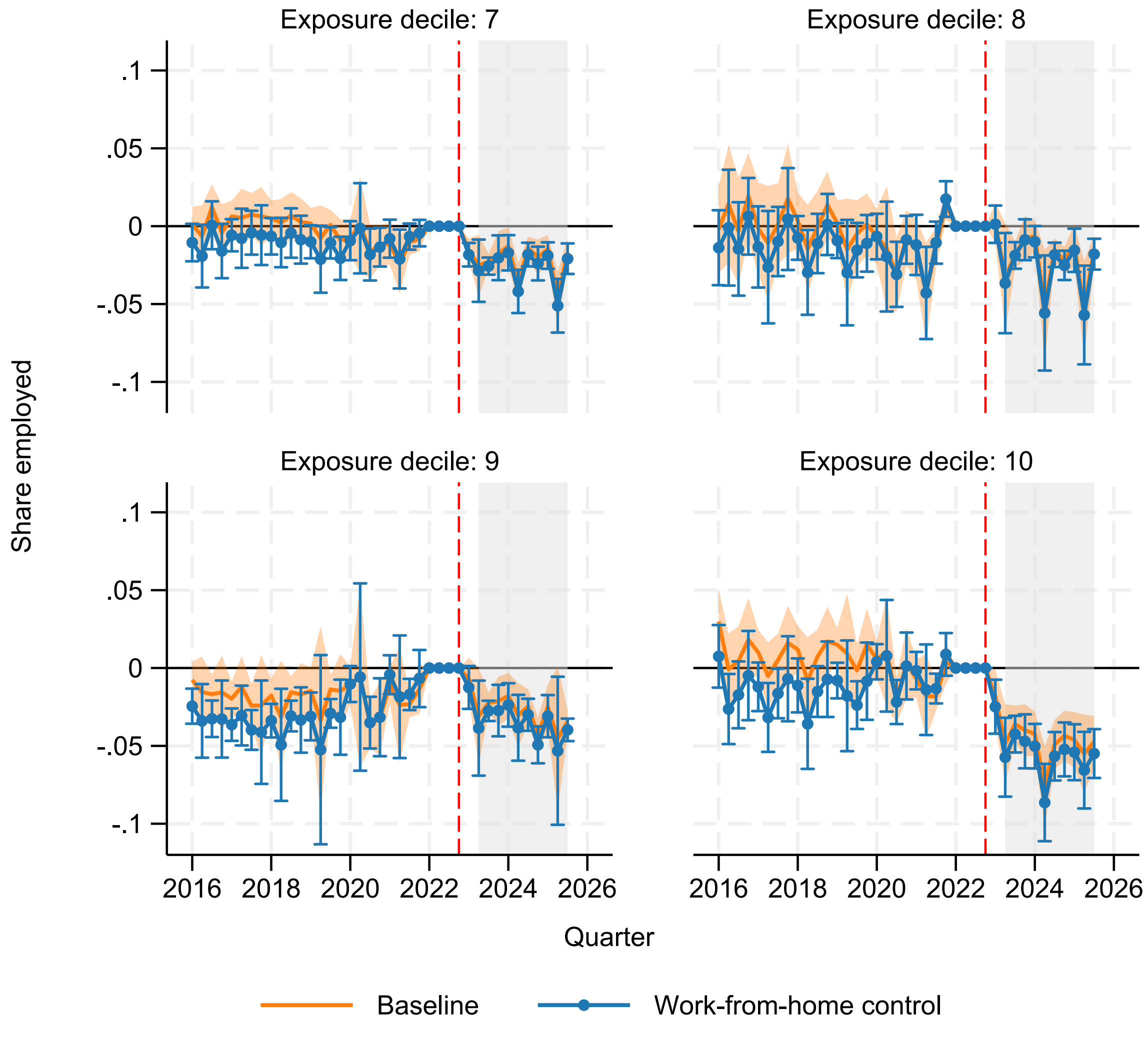
$$y_{it} = \sum_{s \notin 2022} \Big[\beta_s E(m(i)) + \gamma_s W(m(i), 2022)\Big] + \delta_{c(i),m(i)} + \phi_{c(i),t} + \tau_{m(i),q(t)} + \varepsilon_{it}. \quad (4)$$

The coefficients β_s and γ_s represent the marginal effects of a one standard deviation increase in AI exposure or work-from-home rates, respectively.

In Lambert and Schindler, almost all of the negative employment and earnings effects load onto work-from-home. Figures A11 and A12 show that we do not reproduce this finding. Notably, one would expect that the estimated effect of AI and work-from-home should be near zero from 2016 to 2020, since neither was prevalent in this time frame. Yet, we estimate large, offsetting coefficients

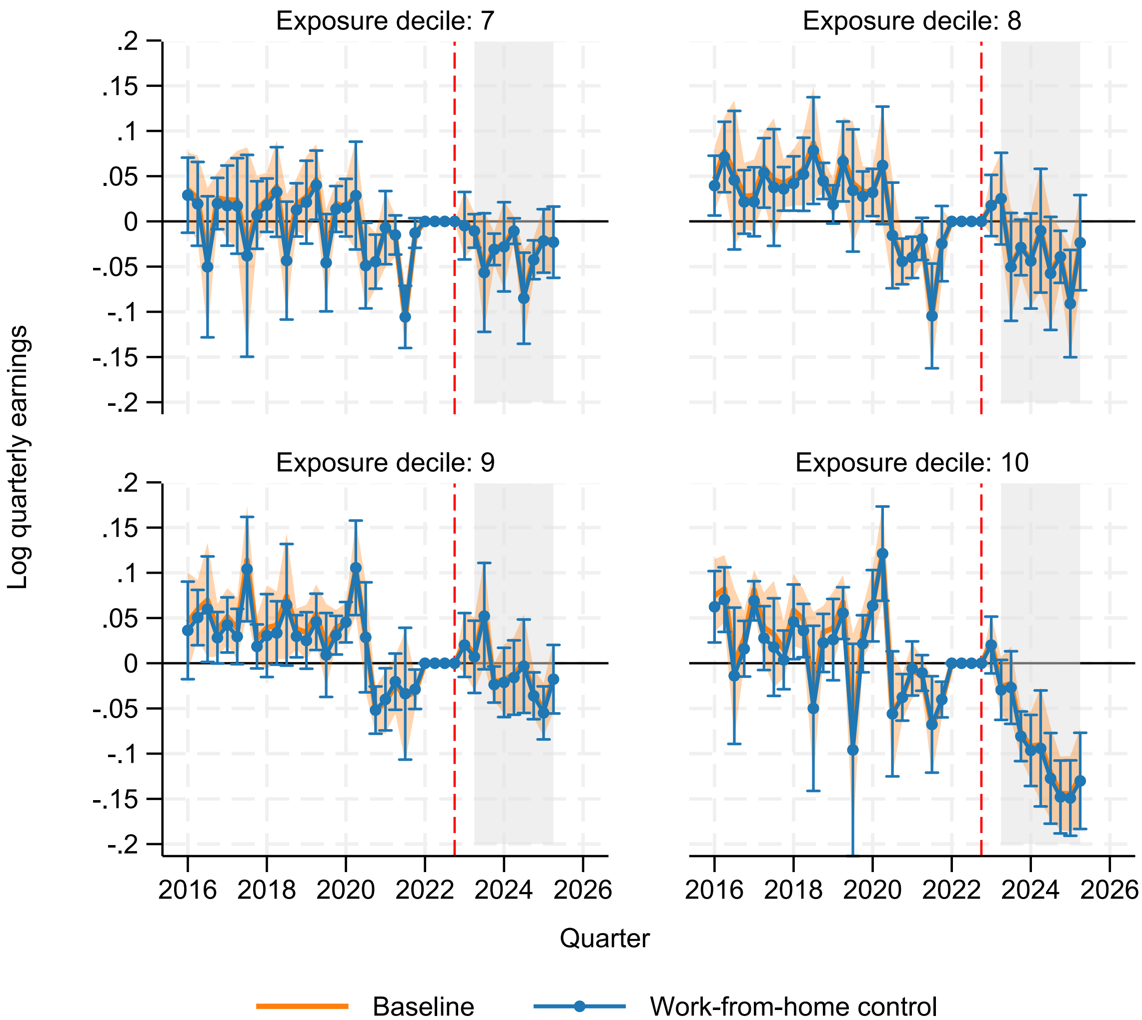
and state.

Figure A9: Initial employment with time varying work-from-home control



Notes: Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Coefficients from regression with time varying work-from-home rates included as controls in addition to institution-major, institution-quarter, and major-quarter of year fixed effects. Standard errors are clustered by college major. Error bars denote 95% confidence intervals. Baseline model estimates and confidence intervals without work-from-home controls are plotted in light blue. Quarter denotes observation period. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 5,685,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

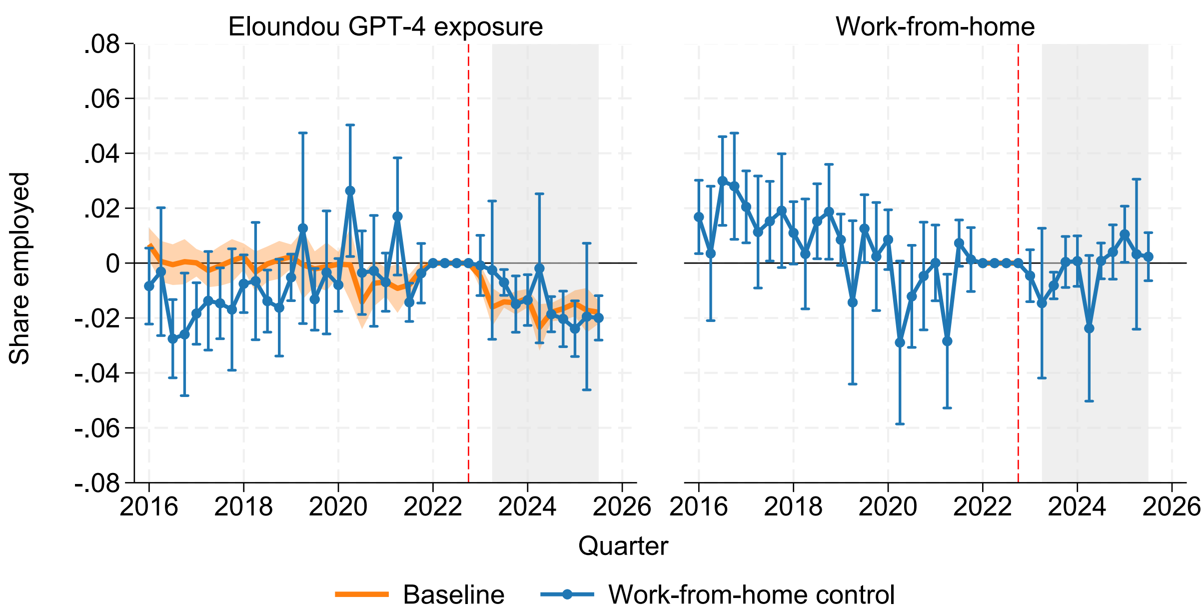
Figure A10: Initial earnings with time varying work-from-home control



Notes: Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. Coefficients from regression with time varying work-from-home rates included as controls in addition to institution-major, institution-quarter, and major-quarter of year fixed effects. Standard errors are clustered by college major. Error bars denote 95% confidence intervals. Baseline model estimates and confidence intervals without work-from-home controls are plotted in light blue. Quarter denotes observation period. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 3,225,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

prior to the COVID pandemic, with widened standard errors consistent with collinearity issues. The estimated effect of AI exposure on initial employment following the release of ChatGPT (Figure A11) is consistently negative and similar to the effect of AI exposure in a restricted model where work-from-home is excluded (shown in shaded blue). The coefficients on work-from-home are predominantly near zero, and if anything, trending upward. When initial quarterly earnings are the outcome (Figure A12), the post-period coefficients on both explanatory variables are statistically indistinguishable from zero. Note that pre-period estimates on the AI exposure covariate are above zero, so the post-period coefficients are still a departure from the long-run average. For work-from-home, the post-period coefficients on earnings are little changed in comparison to the pre-period.

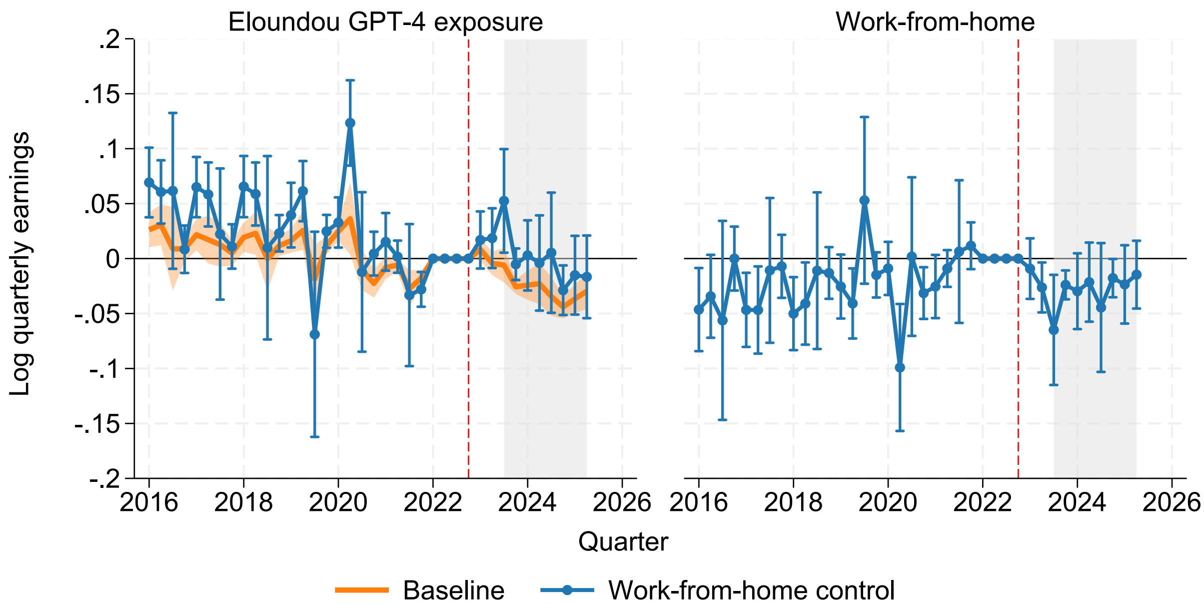
Figure A11: Initial employment with continuous exposure and 2022 work-from-home



Notes: Employment share calculated as the fraction of graduates with non-zero earnings for the quarter. Coefficient from regression with Eloundou et al. GPT-4 Beta exposure (left panel) and 2022 work-from-home rates (right panel) treated as continuous variables in addition to institution-major, institution-quarter, and major-quarter of year fixed effects. Standard errors are clustered by college major. Error bars denote 95% confidence intervals. A restricted model with only GPT-4 Beta exposure, in addition to the fixed effects, plotted in light blue with 95% confidence interval range. Quarter denotes observation period. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 5,685,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

It is unclear how much the specification of Equation 4 can really tell us. As previously mentioned, both variables are highly correlated, more so in our setting than in Lambert and Schindler's, and a model that exploits neither time nor spatial variation may be unable to reliably distinguish

Figure A12: Initial earnings with continuous exposure and 2022 work-from-home



Notes: Log quarterly earnings based on earnings reported for the dominant job if the worker had positive earnings in that same job in both the preceding and subsequent quarters. Coefficient from regression with Eloundou et al. GPT-4 Beta exposure (left panel) and 2022 work-from-home rates (right panel) treated as continuous variables in addition to institution-major, institution-quarter, and major-quarter of year fixed effects. Standard errors are clustered by college major. Error bars denote 95% confidence intervals. A restricted model with only GPT-4 Beta exposure, in addition to the fixed effects, plotted in light blue with 95% confidence interval range. Quarter denotes observation period. November 2022, corresponding to the release of ChatGPT-3.5, marked with a red line. The gray shaded region identifies cohorts who graduated after the release of ChatGPT-3.5. Number of observations: 3,225,000. All results were approved for release by the U.S. Census Bureau, authorization number CBDRB-FY26-CES011-011.

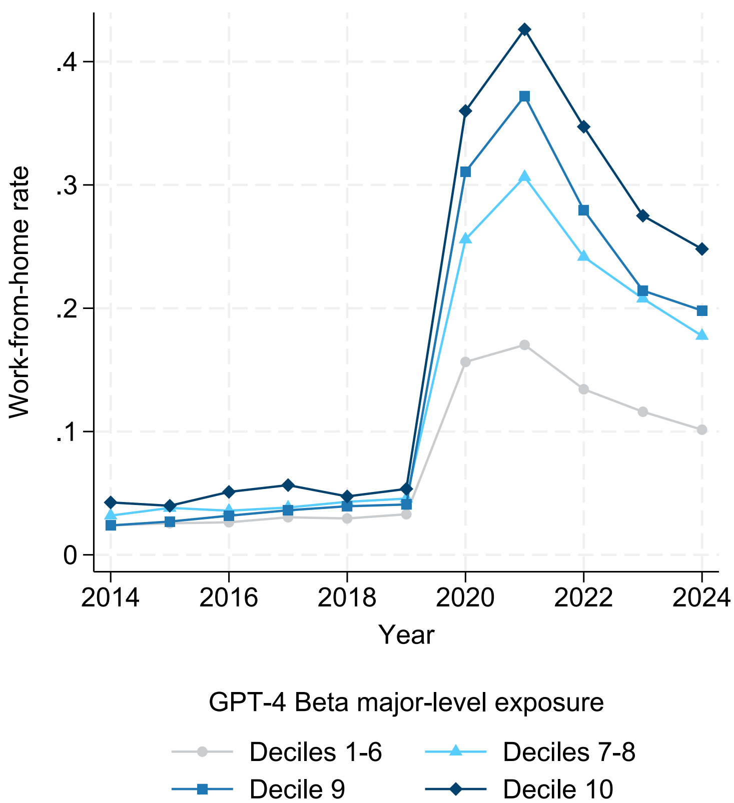
among the two mechanisms. Additionally, we have shown the relationship between AI and labor outcomes to be non-linear with negative effects concentrated among the most exposed majors. Treating exposure as a continuous variable with a linear relationship across all majors likely understates effects for the top decile. Finally, as referenced previously, the coefficients from a continuous difference-in-differences model are weighted averages of causal responses which cannot be interpreted as the global average causal response (Callaway et al., Forthcoming). The weights push the coefficients away from the global average causal response in different ways for each covariate.

Another concern with a work-from-home explanation for the labor market declines among the most AI-exposed college graduates is that the pattern of changes in remote work simply does not align with the pattern of those declines. As shown in Appendix Figure A13, the rate of work-from-home for the most exposed decile of majors early in their career is higher than that of all other exposure deciles prior to the COVID-19 pandemic, increases during the pandemic faster than other exposure deciles, peaks in 2021 above other exposure deciles, and decreases through 2024 by more than all but the ninth exposure decile.[35] If work-from-home was the cause of worsening labor market conditions, we would expect outcomes to improve from 2022 to 2023 as work-from-home rates declined, and to improve more for higher exposure deciles as their work-from-home rate declines were greater. Instead, we observe worsening outcomes following 2022 concentrated among majors most exposed to AI.

In sum, we do not replicate the patterns of Lambert and Schindler (2026), although it is not obvious that we should expect to do so in our particular setting. Previous studies look at workers as a whole, or entry-level workers, or firm-level hiring rates, but we are the first to restrict our analysis exclusively to new college graduates. Published hypotheses as to why work-from-home would depress hiring–such as the higher cost of training in a remote setting–could look very different for recent graduates, whether mitigated because of their specialized training, or exacerbated by their limited work experience. Alternatively, the negative effects of remote work and AI on labor demand could be complementary forces. For instance, both work-from-home and AI might increase the costs of monitoring worker effort or learning on-the-job, and the existence of one technology might magnify the impacts of the other.

[35]The fact that work-from-home rates were highest for decile ten pre-COVID pandemic and increased the most during the pandemic might suggest the occupations these majors work in are the least costly for firms to allow to work remotely. If so, this is another reason to expect work-from-home to have fewer detrimental effects on the most AI-exposed graduates than lesser exposed graduates.

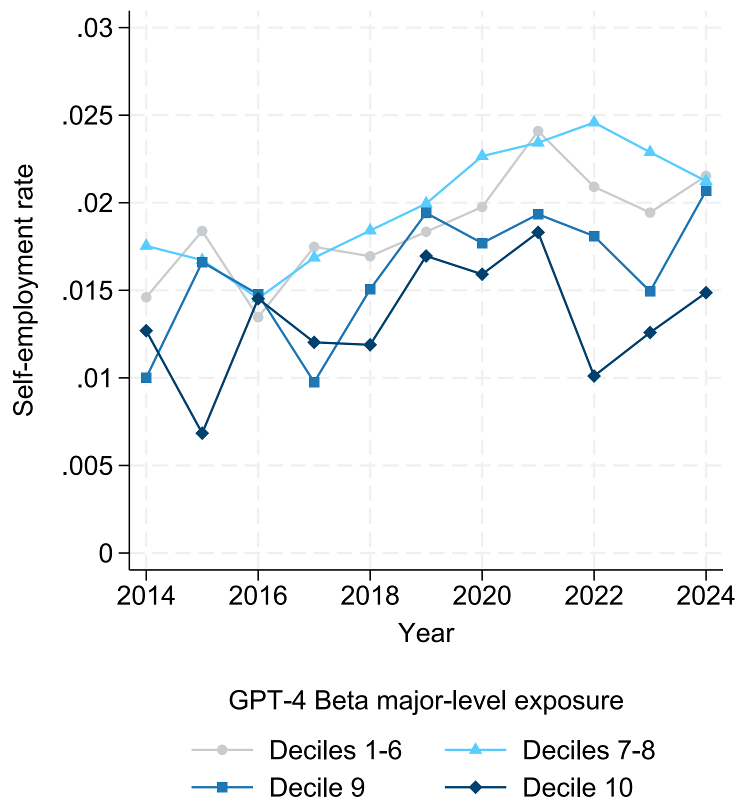
Figure A13: Work-from-home over time



Notes: Work-from-home rates are the fraction of employed bachelor's degree holders aged 21-29 in the ACS that answered the question "how did [you] usually get to work last week" with "worked from home", aggregated by major and weighted by ACS person weights. Separate plots are presented for Eloundou et al. GPT-4 Beta exposure deciles with deciles one through six and seven through eight combined together. Number of observations in ACS: 832,282.

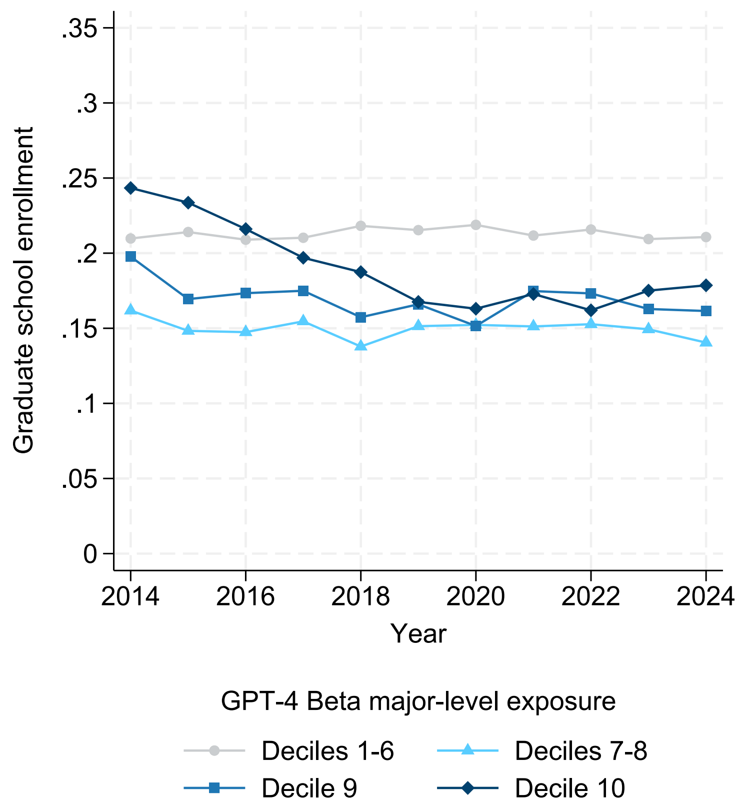
D Alternative economic activities

Figure A14: Self-employment rates over time



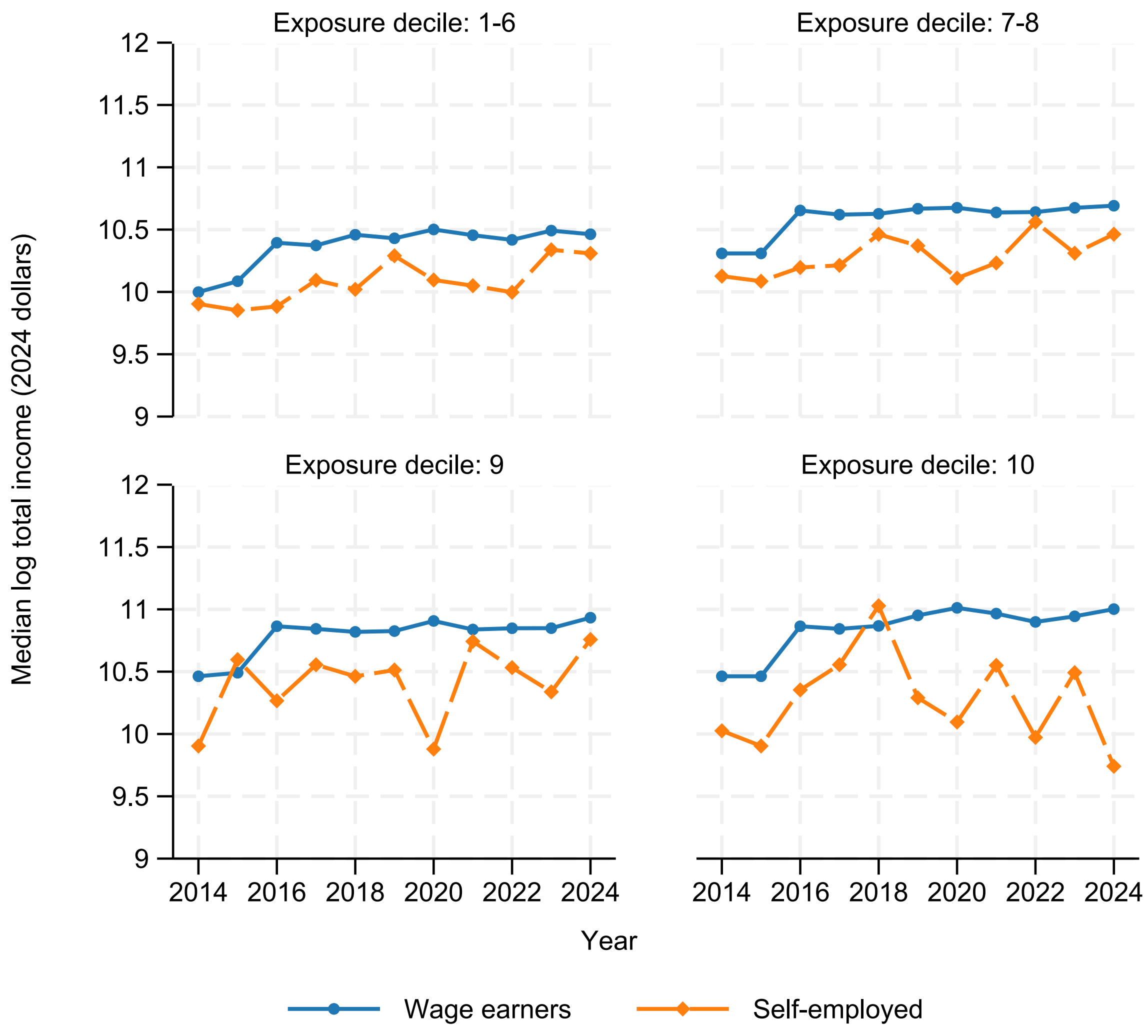
Notes: Self-employed rates are the fraction of bachelor's degree holders aged 21-24 in the ACS that answered the question "Which one of the following best describes this [your] employment last week or the most recent employment in the past 5 years?" with "Owner of [incorporated/non-incorporated] business, professional practice, or farm", aggregated by major and weighted by ACS person weights. Separate plots are presented for Eloundou et al. GPT-4 Beta exposure deciles with deciles one through six and seven through eight combined together. Number of observations in ACS: 309,468.

Figure A15: Graduate school enrollment over time



Notes: Graduate school enrollment rates are the fraction of bachelor's degree holders aged 21-24 in the ACS that answered the question "At any time in the last 3 months, [have you] attended school or college?" with "Yes" and "What grade or level was this person attending?" with "Graduate or professional school beyond a bachelor's degree", aggregated by major and weighted by ACS person weights. Separate plots are presented for Eloundou et al. GPT-4 Beta exposure deciles with deciles one through six and seven through eight combined together. Number of observations in ACS: 309,468.

Figure A16: Median total income by employment type over time



Notes: Total income is the value provided to the question "What was [your] total income during the past 12 months?" among bachelor's degree holders aged 21-24 who are currently employed and worked at least 40 weeks over the last year. Income is adjusted to 2024 dollars using the CPI-U not-seasonally adjusted series. Income reported separately for "wage earners" – those who answered the question, "Which one of the following best describes this [your] employment last week or the most recent employment in the past 5 years?" with an answer in either the "private sector employee" or "government employee" categories – and "self-employed" – those who answered the same question with, "'Owner of [incorporated/non-incorporated] business, professional practice, or farm". Number of observations in ACS: 181,840.